

Design, Implementation, Verification, and Performance Evaluation of CQI-Based BLER-Constrained Adaptive Modulation and Coding for 8×8 and 16×16 MIMO-OFDM Links

Md Moklesur Rahman

Independent Researcher, Finland | moklesur.eee@gmail.com | github.com/dipucwc

Abstract

The threshold-based adaptive modulation of the companion baseline study selects the QAM order from the configured operating SNR and keeps the coding rate fixed at one half, so the transmitter never observes the link quality that the receiver actually experiences and cannot trade coding protection against throughput. This work replaces that open-loop rule with a receiver-driven, BLER-constrained adaptive modulation and coding (AMC) chain for the verified 8×8 and 16×16 MIMO-OFDM platform. The receiver estimates per-stream, per-subcarrier post-equalization SINR from the estimated channel and the deployed unbiased MMSE detector, maps the resource-element SINRs to a codeword-level effective SINR with the exponential effective-SINR mapping (EESM) alongside mean and minimum baselines, generates a channel-quality indicator (CQI) as the highest entry of a nine-level simplified research MCS table whose calibrated BLER prediction satisfies a 0.10 target, and transmits CRC-16-protected codewords at the selected modulation order and punctured coding rate (1/2, 2/3, 3/4 of the constraint-length-7 mother code). Block success is defined solely by the CRC. The practical controller applies the CQI through a one-slot feedback delay, an ideal same-slot variant provides the causality-free upper bound, a perfect pilot-time-CSI reference bounds the reporting loss, and an outer-loop link-adaptation (OLLA) offset driven by CRC feedback is evaluated as an advanced correction. Per-mapping BLER-versus-effective-SINR curves are calibrated through the full MIMO link with adaptive Monte Carlo stopping, and all adaptation methods are compared on identical channel and noise realizations against fixed-MCS and threshold baselines. Every quantitative finding is reported directly from the executed simulation files. In the executed runs the practical delayed-EESM controller reached the 0.10 target at 12 dB SNR on both arrays and held the measured BLER at or below the target at every sweep point from there to 30 dB, with the worst compliant value occurring at the crossing itself, 0.075 on the 8×8 array and 0.034 on the 16×16 array, and compliant-region averages of 0.025 and 0.014. At 30 dB it delivered 52341 and 106457 CRC-qualified bits per slot, a factor of 1.98 and 2.01 above the fixed-MCS baseline and 9.9 and 30.0 percent above the Project 1 threshold rule, which violated the target across nearly the entire sweep on both arrays. The one-slot feedback delay cost 333 bits per slot at 8×8 and 1.25 percent at 16×16 against the causality-free same-slot bound, and the practical controller finished within 1.0 percent of the pilot-time-CSI reference at 8×8 and within 1.4 percent at 16×16. The outer loop tracked the 0.10 target from below on both arrays, and on the 16×16 array its terminal goodput of 125860 bits per slot exceeded the pilot-time-CSI reference, which the reference bounds only the reporting loss and not the full link permits.

Keywords: MIMO-OFDM, link adaptation, channel quality indicator, adaptive modulation and coding, effective SINR, EESM, block error rate, MCS selection, outer-loop link adaptation, goodput

1. Introduction

Section 1 states the motivation and scope of the study. The companion baseline established a verified 8×8 and 16×16 MIMO-OFDM physical-layer platform with comb-pilot Wiener channel estimation, unbiased soft-output MMSE detection, and a threshold-based adaptive-modulation rule with fixed rate- $1/2$ convolutional coding. That adaptation rule has a structural limitation that motivates the present work: the modulation order is chosen from the configured operating SNR, a transmitter-side constant, while the quantity that determines decoding success is the post-equalization SINR that the receiver experiences after channel-estimation error, residual inter-stream interference, and equalizer noise enhancement. At one configured SNR the per-stream, per-subcarrier SINR of a frequency-selective MIMO link spans a wide range, and a single threshold on the configured SNR does not observe that variation.

Four further limitations follow. A fixed coding rate wastes protection when the link is strong and provides too little when it is weak, so modulation and coding rate must adapt together, which is the defining property of AMC. The baseline measured a hard-decision coded goodput proxy from bit errors, whereas link adaptation should be driven by the delivery of whole blocks; the block error rate connects directly to retransmissions and delivered payload and requires a CRC-qualified definition of block success. A practical receiver reports its channel quality over a feedback channel, so the report that selects the MCS of a slot was measured at least one slot earlier, and this delay creates adaptation errors on a time-varying channel. Per-layer and per-subcarrier SINR variation means that a single wideband MCS decision needs a principled scalar summary of the codeword quality, which is the effective-SINR mapping problem.

1.1 Contribution and organization

This work addresses the four limitations on the verified platform. The contribution is not the invention of CQI, EESM, or OLLA, which are established techniques, but a complete, executable, verified implementation and a controlled comparison in which every method sees identical channel and noise realizations and every reported number traces to a stored result file. The remainder of the report is organized as follows. Section 2 reviews the related techniques. Section 3 summarizes the baseline system model. Section 4 derives the post-equalization link-quality estimation. Section 5 develops the effective-SINR mappings. Section 6 defines the CQI and the MCS table. Section 7 presents the variable-rate coding chain and the CRC-based BLER measurement. Section 8 states the complete algorithms. Section 9 documents the MATLAB implementation. Section 10 defines the verification methodology. Section 11 lists the simulation parameters. Section 12 reports the executed results. Section 13 states the limitations, and Section 14 concludes.

2. Related Work

Section 2 places the study among established link-adaptation techniques. Adaptive modulation for fading channels was formalized by Goldsmith and Chua [1], who showed that matching the constellation to the instantaneous channel state approaches the capacity of the fading link, and the joint adaptation of modulation and coding rate under an error-rate constraint became the standard link-adaptation architecture of HSPA, LTE, and NR [2], [3]. In that architecture the receiver reports a channel-quality indicator, the scheduler maps the CQI to a modulation-and-coding-scheme index, and the target is a first-transmission BLER, conventionally ten percent, that balances throughput against retransmission delay [3], [4].

2.1 Effective-SINR mapping

Reducing the per-resource-element quality of a wideband codeword to one scalar is the link-to-system mapping problem. The exponential effective-SINR mapping was introduced for system-level evaluation of OFDM links [5], [6]: a per-MCS calibration parameter makes the AWGN BLER curve at the effective SINR predict the BLER of the frequency-selective link. Mean-SINR mapping overestimates the quality of selective codewords because decoding is dominated by the weakest resource elements, and minimum-SINR mapping is over-conservative; EESM sits between the two and is the de facto research standard alongside mutual-information-based mappings [6], [7].

2.2 Outer-loop adaptation and feedback effects

Outer-loop link adaptation corrects the residual bias of any inner-loop prediction by nudging an SINR offset after every ACK or NACK so that the long-run BLER converges to the target [8]. Feedback delay and CQI reporting errors are known to erode adaptation gains on time-varying channels [4], [9], and MIMO link adaptation adds per-stream quality variation to the mapping problem [7], [10]. This work claims no novelty in any of these techniques individually; the contribution is the complete verified implementation and the controlled comparison on the verified baseline platform.

3. Baseline System Model

Section 3 summarizes the platform, which is the verified baseline of the companion study [11]. One slot consists of one comb-pilot OFDM symbol followed by thirteen data OFDM symbols on a 256-subcarrier grid with a 20-sample cyclic prefix at a 30.72 MHz sample rate. The channel is a five-tap TDL-A power-delay profile with independent Rayleigh tap matrices per slot and a simplified common-Doppler phase-evolution model across the slot at 30 km/h and 3.5 GHz. Each transmit antenna owns a non-overlapping comb of pilot subcarriers; the receiver forms least-squares estimates at the pilots and interpolates all subcarriers with a Wiener MMSE filter built from the tap-profile correlation matrices at the operating noise variance. Equal total transmit power is maintained by scaling the channel by $1/\sqrt{N_t}$, and the detector is the regularized MMSE equalizer whose gain-corrected unbiased output feeds every metric and both decoding paths of the baseline study. The baseline adaptation, retained here as a comparison method, selects QPSK below 8 dB configured SNR, 16-QAM from 8 dB to below 18 dB, and 64-QAM from 18 dB, always at coding rate one half. The full derivations are in the baseline report [11] and are not repeated; the present study changes nothing in this platform and adds the adaptation chain on top of it.

4. Post-Equalization Link-Quality Estimation

Section 4 derives the SINR that each spatial stream experiences at the output of the deployed detector, computed by the receiver from the Wiener channel estimate and the known noise variance. Let $\hat{H}k$ denote the estimated channel matrix at subcarrier k after the transmit-power scaling by $1/\sqrt{N_t}$, and let σ^2 be the receiver-noise variance. The detector in use is the regularized MMSE equalizer of the baseline platform,

$$Wk = (\hat{H}k^H \hat{H}k + \sigma^2 I)^{-1} \hat{H}k^H \quad (1)$$

whose rows act on the received vector. With $Gk = Wk\hat{H}k$, the diagonal entry $Gk(l,l)$ is the desired-stream gain of stream l , the off-diagonal entries of row l are the residual inter-stream leakage after equalization, and the row energy of Wk measures the noise enhancement of the equalizer. For transmitted symbol energy E_s , the receiver-side per-stream SINR estimate at subcarrier k is

$$\text{SINR}(l,k) = |G_k(l,l)|^2 E_s / (\sum_{j \neq l} |G_k(l,j)|^2 E_s + \|w_l\|^2 \sigma^2) \quad (2)$$

in which $G_k(l,j)$ is the entry of G_k in row l and column j , w_l is row l of W_k , E_s is the transmitted symbol energy, and σ^2 is the receiver-noise variance. The numerator is the power of the desired stream at the equalizer output; the two denominator terms are the residual inter-stream interference and the enhanced thermal noise. Because the weights in equation (1) are exactly the weights of the deployed detector, equation (2) describes the detector actually in use rather than an idealized receiver.

4.1 Slot-level SINR and the pilot-time reference

The channel estimate is formed once per slot at the pilot symbol, so the SINR estimate of equation (2) is one value per stream and subcarrier that the controller holds constant across the thirteen data symbols of the slot. The practical receiver cannot re-estimate the channel inside the slot, and the resulting mismatch against the Doppler-evolved data-symbol channel is part of the estimation error that the delayed-feedback results expose. For verification and for the reference method labeled oracle, a reference SINR is computed by evaluating equations (1) and (2) on the true pilot-time channel instead of the estimate. The data detector continues to use the estimated-channel weights in every method, and the calibrated curves are built on the estimated-SINR axis, so this reference provides perfect pilot-time channel knowledge for CQI generation rather than a strict full-link upper bound. The verification requirement is consistent trends and reasonable average agreement, which Section 12.3 reports through the overlap of the estimated-CQI and reference-CQI curves.

5. Effective-SINR Mapping

Section 5 develops the scalar quality summary. One codeword occupies every data subcarrier, every data OFDM symbol, and one spatial layer. The corrected controller evaluates its mappings on the per-layer resource-element SINR grid of equation (2) and forms one effective SINR per codeword, and a common-MCS rule over the per-layer predictions produces the wideband decision. Three mappings are implemented. The arithmetic mean of the linear SINR values,

$$\text{SINR_mean} = (1/N) \sum_n \text{SINR}_n \quad (3)$$

where N is the number of resource elements of the evaluated grid and SINR_n the linear SINR of resource element n , weights every resource element equally. It overestimates the quality of frequency-selective codewords because convolutional decoding is dominated by the weakest stretches of the codeword, which the mean averages away. The minimum mapping,

$$\text{SINR_min} = \min(n) \text{SINR}_n \quad (4)$$

predicts from the single worst resource element and therefore selects an unnecessarily low MCS whenever one deep notch exists. The primary mapping is the exponential effective-SINR mapping,

$$\text{SINR_eff} = -\beta_m \ln((1/N) \sum_n \exp(-\text{SINR}_n / \beta_m)) \quad (5)$$

in which β_m is the calibration parameter of MCS m and the sum runs over the N resource elements of the evaluated grid. The exponential kernel weights weak resource elements the way the decoder feels them: as β_m grows the mapping approaches the mean, and as β_m shrinks it approaches the minimum, so one parameter per MCS places the mapping between the two baselines. The β_m values are configured calibration parameters listed in Section 11; the predicted-versus-measured comparison of Section 12 validates them against the executed curves.

The controller evaluates equation (5) once per candidate MCS with that candidate's own β_m , which is the standard EESM usage. Each mapping is calibrated against its own BLER curves so that every controller queries the curves that match its quality axis.

6. CQI and MCS Design

Section 6 defines the decision table and the selection rule. Table 1 lists the simplified research MCS table: the three QAM orders of the baseline platform crossed with three convolutional coding rates. This is a custom research table constructed for this study; it is not a standardized 3GPP MCS table, and no standards compliance is claimed.

Table 1. Simplified research MCS table.

MCS index	Modulation	Order	Bits/symbol	Coding rate	Nominal SE (bit/symbol)
0	QPSK	4	2	1/2	1.00
1	QPSK	4	2	2/3	1.33
2	QPSK	4	2	3/4	1.50
3	16-QAM	16	4	1/2	2.00
4	16-QAM	16	4	2/3	2.67
5	16-QAM	16	4	3/4	3.00
6	64-QAM	64	6	1/2	3.00
7	64-QAM	64	6	2/3	4.00
8	64-QAM	64	6	3/4	4.50

The first column of Table 1 lists the MCS index, the second to fourth columns the constellation and its order and bits per symbol, the fifth column the coding rate produced by puncturing, and the sixth column the nominal spectral efficiency, the product of bits per symbol and coding rate,

$$\eta_m = b_m R_m \quad (6)$$

where b_m is the bits per symbol and R_m the coding rate of MCS m . The nominal spectral efficiency never decreases with the index, which the verification gates check explicitly; entries 5 and 6 share the value 3.00 bit/symbol with different modulation-coding trade-offs, and the table orders 64-QAM rate-1/2 above 16-QAM rate-3/4 because its higher-order constellation offers more headroom at high SINR. The CQI is defined as the highest MCS whose predicted BLER at the reported effective SINR satisfies the target,

$$CQI = \max \{ m : \text{BLER}_m(\text{SINR}_{\text{eff},m}) \leq \text{BLER}_{\text{target}} \} \quad (7)$$

in which BLER_m is the calibrated-curve prediction of Section 7 evaluated at the effective SINR computed with that candidate's β_m , and the simplified research model treats the CQI and the MCS index as equal. The transmitter selection applies the same rule to the report it possesses,

$$m^* = \max \{ m : \text{BLER}_m(\text{SINR}_{\text{eff},m} + \Delta_{\text{OLLA}}) \leq \text{BLER}_{\text{target}} \} \quad (8)$$

where Δ_{OLLA} is the outer-loop offset of Section 8, zero for the methods without the outer loop. Boundary handling is explicit: when no candidate satisfies the target the selection falls back to MCS 0, when the top entry satisfies it the selection saturates at MCS 8, and the selector records for every decision whether the target was met, the fallback was taken, the table top was reached, or no valid prediction existed. When two admissible entries share a nominal spectral efficiency the selector keeps the higher-index entry, and a spectral-efficiency-first tie-break is available in configuration.

7. Variable-Rate Channel Coding and BLER Measurement

Section 7 states the coding chain and the block-error definition. The mother code is the constraint-length-7, rate-1/2 convolutional code with generator polynomials (133, 171) in octal, the code of the baseline platform. Higher rates are produced by puncturing: the rate-2/3 mask transmits three of every four mother bits and the rate-3/4 mask four of every six. At the receiver the punctured positions are restored before decoding as zero-valued log-likelihood-ratio erasures, and decoding runs the terminated soft-decision Viterbi algorithm on the mother-code trellis.

7.1 Codeword sizing and interleaving

Each spatial layer of each slot carries one codeword, so one slot produces N_t block verdicts under one wideband MCS. The codeword container of one layer is the channel-bit capacity of that layer,

$$N_{\text{chan}} = N_{\text{sc}} \cdot N_{\text{sym}} \cdot b_m \quad (9)$$

with $N_{\text{sc}} = 256$ data subcarriers, $N_{\text{sym}} = 13$ data OFDM symbols, and b_m bits per symbol, and the information payload of the codeword is

$$K_{\text{info}} = \lfloor N_{\text{chan}} R_m \rfloor - L_{\text{CRC}} - L_{\text{tail}} \quad (10)$$

where $L_{\text{CRC}} = 16$ is the CRC length and $L_{\text{tail}} = 6$ the tail bits that terminate the encoder in the zero state. Eight entries of Table 1 fill their containers exactly. MCS 4 requires one known zero-padding bit per spatial-layer codeword, because the rate-2/3 puncturing of its mother codeword retains all but one of the container bits; the implementation appends the padding and records its length at run time, which Gate 17 reports. A seeded random bit interleaver spreads each codeword over its container before constellation mapping: without interleaving, the deep notches of the frequency-selective channel produce error bursts that the convolutional decoder cannot absorb, whereas the interleaved codeword experiences its resource elements in randomized order and behaves as the effective-SINR abstraction assumes.

7.2 CRC block success and delivered-payload metrics

The transmitted CRC-16-CCITT field, generator polynomial $x^{16} + x^{12} + x^5 + 1$ with an all-zero initial register, defines block success: a block is delivered if and only if the recomputed remainder over the received information-plus-CRC field is zero. No BER-derived block statistic appears anywhere in the study. The measured block error rate is

$$\text{BLER} = N_{\text{failed}} / N_{\text{total}} \quad (11)$$

with N_{failed} the CRC-failing blocks and N_{total} the transmitted blocks, and every reported BLER carries the 95 percent Wilson score interval

$$\text{CI} = \left(\hat{p} + z^2/2n \pm z \sqrt{\hat{p}(1-\hat{p})/n + z^2/4n^2} \right) / \left(1 + z^2/n \right) \quad (12)$$

where $\hat{p}$ is the measured BLER, n the block count, and $z = 1.96$. Calibration transmits CRC-protected blocks through the full MIMO link at fixed MCS over a per-MCS SNR window and uses adaptive Monte Carlo stopping: a point ends when the configured minimum number of block errors has been observed or the configured maximum number of blocks has been simulated, whichever comes first. Each stored point pairs the measured BLER with the measured mean effective SINR of that point, and prediction between points uses linear interpolation after a running-minimum enforcement of monotonicity. The delivered-payload metrics are the CRC-qualified goodput per slot,

$$G = B_{\text{delivered}} / N_{\text{slots}} \quad (13)$$

where $B_{\text{delivered}}$ counts the information bits of CRC-passing blocks only, and the spectral efficiency normalized per subcarrier and OFDM symbol,

$$SE = B_{\text{delivered}} / (N_{\text{slots}} \cdot N_{\text{sc}} \cdot N_{\text{sym}}) \quad (14)$$

which aggregates the delivered bits of all spatial layers onto the time-frequency grid. The term goodput is used exclusively for these CRC-qualified quantities.

8. Proposed BLER-Constrained AMC Algorithm

Section 8 states the complete adaptation chain as eight numbered procedures. Equations are referenced by their continuous numbers, and the mapping from every procedure to its implementing MATLAB function is the traceability table of Section 9. Two mechanisms are stated formally first. The report that drives the MCS of slot t under the practical delayed methods is the report generated D slots earlier,

$$m(t) = \text{CQI}(t - D) \quad (15)$$

where D is the configured feedback delay in slots and the first D slots use the configured initial MCS because no earlier report exists. The ideal same-slot variant sets $D = 0$ and is presented exclusively as a causality-free upper bound. The outer-loop offset is updated after every CRC verdict,

$$\Delta_{\text{OLLA}} \leftarrow \Delta_{\text{OLLA}} - \Delta_{\text{down}} (\text{failure}), \quad \Delta_{\text{OLLA}} \leftarrow \Delta_{\text{OLLA}} + \Delta_{\text{up}} (\text{success}) \quad (16)$$

with the positive-offset-is-aggressive convention of equation (8), and the two steps obey the target-locked ratio

$$\Delta_{\text{up}} = \Delta_{\text{down}} \cdot p / (1 - p) \quad (17)$$

where p is the target BLER, so failures produce a larger downward correction than the upward adjustment after each success and the offset drift vanishes exactly at the target operating point.

Algorithm 1: Post-equalization SINR calculation

1. For each subcarrier k , scale the estimated channel by $1/\sqrt{N_t}$.
2. Form the regularized MMSE weight W_k of equation (1) and the product $G_k = W_k \hat{H}_k$.
3. For each stream l , compute the SINR estimate of equation (2) from the diagonal gain, the off-diagonal leakage, and the weighted noise power.
4. Return the $N_t \times N_{\text{sc}}$ SINR grid together with the weights, bias gains, and effective noise variances of the detector.

The procedure operates as follows. The routine runs once per slot on the pilot-time Wiener estimate, so its outputs are shared by the detector and the controller: the same weights that equalize the data symbols generate the SINR grid that drives the adaptation, and the grid is held constant across the thirteen data symbols as stated in Section 4.

Algorithm 2: Effective-SINR mapping

1. Collect the resource-element SINR values of the codeword.
2. Under the mean mode, apply equation (3); under the minimum mode, equation (4).
3. Under the EESM mode, apply equation (5) with the calibration parameter β_m of the candidate MCS.
4. Return the effective SINR in decibels.

The procedure operates as follows. The controller calls the mapping once per candidate MCS with that candidate's own β_m , so one SINR grid produces nine candidate-specific effective SINRs per decision under the EESM mode and a single shared value under the mean and minimum baselines.

Algorithm 3: CQI generation

1. For each candidate MCS from lowest to highest, compute the candidate-specific effective SINR (Algorithm 2).
2. Predict the candidate BLER from its calibrated curve with monotone interpolation.
3. Record the highest candidate whose prediction satisfies the target of equation (7).
4. Report that index as the CQI; report zero when no candidate satisfies the target.

The procedure operates as follows. The CQI equals the MCS index in this simplified research model, so the report is directly actionable by the transmitter without a separate mapping table, and the monotonicity of the prediction in the effective SINR guarantees the monotonicity of the CQI, which Gate 7 checks.

Algorithm 4: CQI-to-MCS selection

1. Evaluate the predicted BLER of every table entry at the reported quality, offset-corrected per equation (8).
2. Select the highest entry satisfying the target.
3. Fall back to MCS 0 when no entry satisfies the target; saturate at MCS 8 when the top entry satisfies it.
4. Record the selection status: target met, lowest-MCS fallback, highest-MCS selected, or no valid prediction.

The procedure operates as follows. The corrected chain applies the outer-loop offset at the transmitter after the feedback delay, so the selector operates on the delayed per-layer quality report rather than on a pre-corrected integer, and the candidate-specific quality information is preserved through the delay buffer, which Gates 14 and 16 verify.

Algorithm 5: Variable-rate encoding and soft decoding

1. Compute the payload size of equation (10); draw the information bits and append the CRC-16 field.
2. Append the tail bits, encode with the (133,171) mother code, and puncture with the mask of the selected rate.
3. Interleave with the seeded per-codeword permutation and map to the selected constellation.
4. At the receiver, demap per-bit LLRs with the per-stream effective noise variance, deinterleave, and restore punctured positions as zero-valued erasures.
5. Decode with the terminated soft Viterbi algorithm, strip the tail, and verify the CRC.

The procedure operates as follows. One codeword per spatial layer keeps the block verdicts statistically useful while the wideband MCS of the slot stays one decision, and the per-MCS padding length of Section 7 is recorded so that the container fill is exact for every table entry once the single MCS 4 padding bit is accounted for.

Algorithm 6: Delayed CQI application

1. Initialize a first-in-first-out buffer of length D with the configured initial MCS.
2. At each slot, read the head of the buffer as the MCS-driving report of this slot, per equation (15).
3. Compute the new per-layer quality report of this slot (Algorithm 3) and push it to the tail of the buffer.
4. Under the optional reporting-error experiment, corrupt the pushed report by one step with the configured probability before buffering.

The procedure operates as follows. Causality is structural rather than procedural: the value read at slot t was inserted no later than slot t – D, so no same-slot measurement can influence its own transmission, which Gate 10 confirms on the buffer directly.

Algorithm 7: OLLA update

1. After the CRC verdict of the slot, decrease the offset by Δ_{down} on each block failure, per equation (16).
2. Increase the offset by Δ_{up} on each block success, with the step ratio of equation (17).
3. Clamp the offset to the configured range.
4. Apply the offset additively to the effective SINR of every candidate at the next selection, per equation (8).

The procedure operates as follows. The corrected loop updates on the failure fraction over all block verdicts of the slot rather than on a single any-layer slot verdict, so the step ratio tuned for a 0.10 event probability drives the block error rate toward the target, and Gate 15 confirms zero drift at the target.

Algorithm 8: Complete CQI-based AMC procedure (one slot)

1. Read the delayed report (Algorithm 6) and select the MCS with the offset applied at the transmitter (Algorithm 4).
2. Encode, interleave, and map one codeword per layer at the selected MCS (Algorithm 5).
3. Transmit through the MIMO-OFDM link; receive the comb pilot and form the Wiener estimate.
4. Equalize the data symbols with the unbiased MMSE detector.
5. Estimate the post-equalization SINR grid (Algorithm 1).
6. Generate the new per-layer quality report (Algorithms 2 and 3) and queue it (Algorithm 6).
7. Decode and check the CRC of every layer codeword (Algorithm 5).
8. Update the OLLA offset from the block verdicts where the outer loop is active (Algorithm 7).
9. Accumulate BLER, goodput, CQI, MCS, switching, and OLLA statistics.

The procedure operates as follows. The ordering embodies the feedback constraint: the MCS of the slot is fixed before the slot's own measurements exist, the measurements of the slot influence later slots only through the buffer, and the CRC verdicts close the outer loop.

Figure 1 presents the complete workflow of one slot.

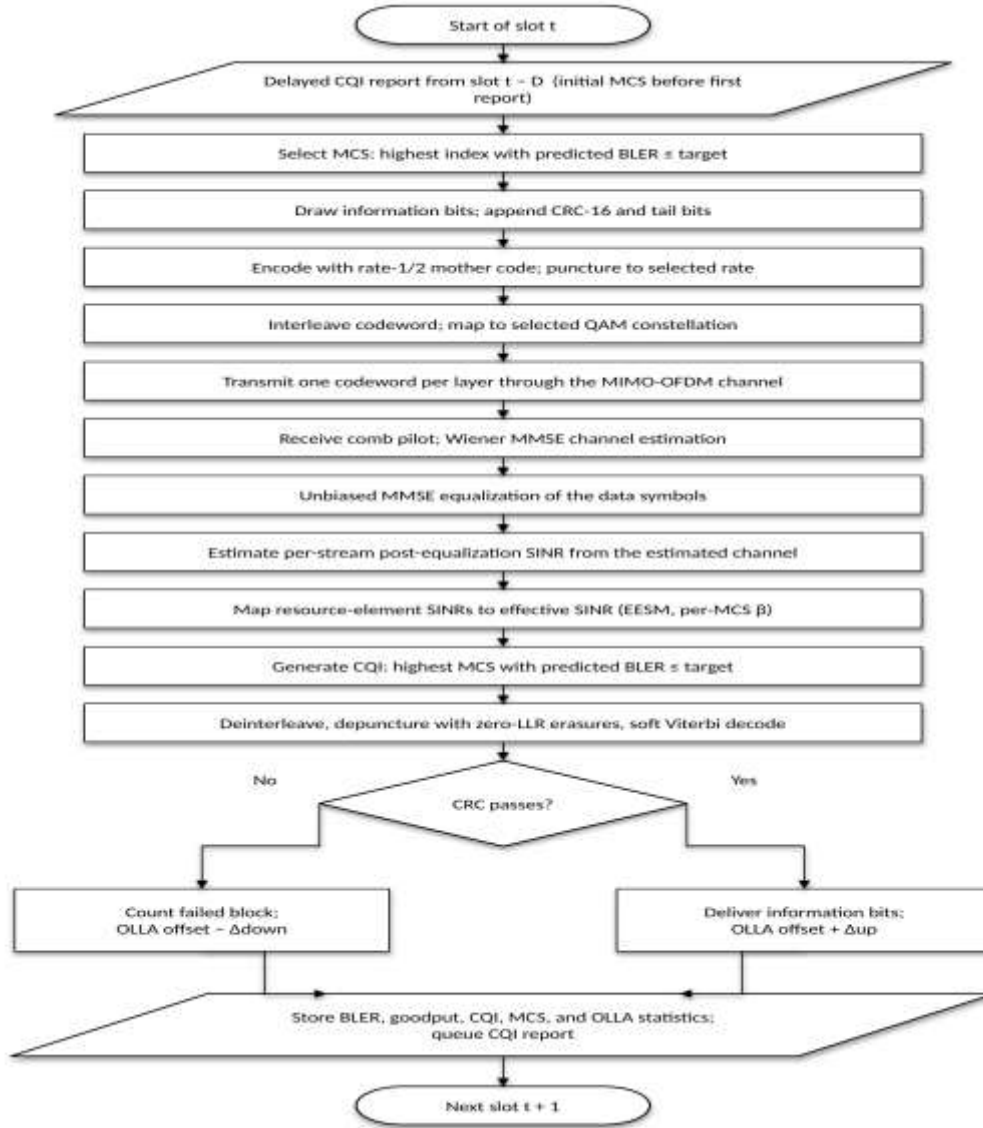


Figure 1. Complete per-slot workflow of the CQI-based BLER-constrained AMC chain.

Figure 1 shows the causal order of one slot: the delayed report enters at the top, the MCS decision precedes every measurement of the slot, the receiver chain from pilot estimation to SINR mapping produces the next report, and the CRC verdict branches into the failed-block and delivered-block paths that update the outer-loop offset before the statistics of the slot are stored.

9. MATLAB Implementation

9.1 File architecture and configuration flow

All configuration lives in `create_cqi_amc_config`, which replicates the verified baseline PHY parameters and adds every controller parameter, so one structure defines a run. The MCS table is built by `create_mcs_table`. The verified baseline functions are reused unchanged for the channel, channel estimation, and detector. Two modes share one engine. In calibration mode, `generate_mcs_bler_curves` transmits fixed-MCS blocks through the full link and writes one curve file per mapping and array, `mcs_bler_curves_eesm`, `mcs_bler_curves_mean`, and `mcs_bler_curves_minimum` for the 8×8 and 16×16 configurations. In online mode, `main_cqi_amc_8x8` and `main_cqi_amc_16x16` call the shared driver `run_cqi_amc_main`, which runs all seven adaptation methods and writes `results_cqi_amc_<array>.csv` with the full per-SNR

column set, `mcs_selection_<array>.csv` with the MCS-specific statistics of the primary method, and `run_config_cqi_<array>.csv` with the run fingerprint that `run_cqi_amc_comparison` validates before any cross-array overlay.

9.2 Shared realizations, seeding, and execution order

The comparability rule of the study is enforced structurally. `prepare_slot_shared` draws one channel evolution, one pilot-noise grid, and one data-noise tensor per slot from a per-slot seeded stream, forms the Wiener estimate, and precomputes the detector weights, bias gains, effective noise variances, and both the estimated and the true-channel SINR grids; `run_slot_link` then transmits any requested MCS through exactly that realization. Every adaptation method of a slot therefore experiences identical channel and noise, and only the payloads differ. The per-slot seed is a deterministic function of the master seed, the SNR index, and the slot index, so the same configuration reproduces the identical run, which Gate 12 confirms. The execution order is `run_verification_gates`; `generate_mcs_bler_curves` for the 8×8 array and then the 16×16 array; `main_cqi_amc_8x8`; `main_cqi_amc_16x16`; `run_cqi_amc_comparison`.

9.3 Algorithm-to-code traceability

Table 2 maps each numbered algorithm and its defining equations to the implementing MATLAB functions.

Algorithm	Equations	Implementing functions
1 Post-equalization SINR	(1)–(2)	<code>estimate_post_eq_sinr</code>
2 Effective-SINR mapping	(3)–(5)	<code>calculate_effective_sinr</code>
3 CQI generation	(7)	<code>estimate_cqi</code> , <code>interpolate_bler_curve</code> , <code>estimate_quality_report</code>
4 MCS selection	(8)	<code>select_mcs_for_target_bler</code>
5 Variable-rate coding	(9)–(10)	<code>append_crc</code> , <code>check_crc</code> , <code>crc16_bits</code> , <code>select_puncturing_pattern</code> , <code>apply_puncturing</code> , <code>depuncture_llr</code> , <code>encode_variable_rate</code> , <code>decode_variable_rate</code>
6 Delayed CQI	(15)	<code>apply_cqi_feedback_delay</code>
7 OLLA update	(16)–(17)	<code>update_olla_offset</code>
8 Complete AMC slot	(13)–(14)	<code>run_cqi_amc_main</code> , <code>prepare_slot_shared</code> , <code>run_slot_link</code> , <code>compute_bler</code>
Calibration	(11)–(12)	<code>generate_mcs_bler_curves</code> , <code>build_link_precompute</code> , <code>build_wiener_filter</code> , <code>verify_curve_mapping</code>

The first column of Table 2 lists the numbered procedure, the second column its defining equations by their continuous numbers, and the third column the MATLAB functions that implement it, so that every mathematical statement of Sections 4 to 8 has a named executable counterpart in the package.

10. Verification Methodology

Section 10 defines the executable verification. Every claim of the study rests on twenty verification gates implemented in `run_verification_gates`, which prints an explicit pass or fail verdict per gate. Gates 1 to 12 cover the coding chain and inner loop: CRC pass and corrupted-block detection, noiseless round trips at rates 1/2, 2/3, and 3/4, puncture and depuncture positions, MCS-table integrity with the nondecreasing spectral-efficiency condition, SINR-estimator gain and noise behavior, effective-SINR and BLER-curve and CQI monotonicity, highest-

satisfying-MCS selection, feedback causality on the delay buffer, outer-loop direction and step asymmetry, and reproducibility under a fixed seed. Gates 13 to 20 cover the corrected architecture: mapping-matched curve tables, the per-layer quality report, zero OLLA drift at the target, transmitter-side offset application after the delay, per-MCS padding-length reporting, nondecreasing CQI on the executed real curves, the reference driven by mapping-matched curves, and Wilson-interval correctness. Table 3 collects the gates; in keeping with the rule that no test is marked as passed until MATLAB confirms it, the executed-result column is completed only from the executed run.

Verification gate	Expected result	Executed result
1 CRC valid / corrupted block	Pass / detected fail	PASS
2 Round trip rates 1/2, 2/3, 3/4	Zero errors	PASS
3 Puncture / depuncture positions	Erasures at removed positions	PASS
4 MCS-table integrity	All checks hold	PASS
5 SINR-estimator behavior	Gain up, noise down	PASS
6 Effective-SINR monotonicity	Nondecreasing	PASS
7 CQI monotonicity	Nondecreasing CQI	PASS
8 BLER-curve monotonicity	Nonincreasing prediction	PASS
9 BLER-target selector	Highest satisfying MCS	PASS
10 Feedback causality	No same-slot use	PASS
11 OLLA directions	Failure down, success up	PASS
12 Reproducibility	Identical repeat run	PASS
13 Mapping-matched curve tables	Curves match mapping	PASS
14 Per-layer quality report	Per-layer report formed	PASS
15 OLLA drift at target BLER	Zero drift at target	PASS
16 OLLA applied after delay	Transmitter-side offset	PASS
17 Per-MCS padding lengths	Reported [0 0 0 0 1 0 0 0]	PASS
18 Nondecreasing CQI on real curves	Nondecreasing	PASS
19 Reference uses matched curves	Mapping-matched	PASS
20 Wilson interval correctness	Correct interval	PASS

The first column of Table 3 names the gate, the second column the expected outcome that defines a pass, and the third column the outcome actually printed by `run_verification_gates`, transcribed from the executed console output, which reported twenty gates passed and zero failed.

11. Simulation Parameters

Section 11 lists the complete parameter set. Table 4 records every configured aspect of the executed run.

Parameter	Value
Antenna configurations	8×8 and 16×16
FFT size / cyclic prefix	256 subcarriers / 20 samples
Data OFDM symbols per slot	13 (one comb-pilot symbol per slot)

Parameter	Value
Channel profile	5-tap TDL-A, delays [0 1 2 4 6] samples, powers [0 -2.2 -4.0 -6.5 -9.0] dB
Doppler model	Simplified common-Doppler phase evolution, 30 km/h at 3.5 GHz
Online SNR range	0 to 30 dB in 2 dB steps
Slots per SNR point	100
Target BLER	0.10
CQI delay	1 slot (delayed methods); 0 for the ideal bound and the pilot-time-CSI reference
CQI error probability	0 (optional experiment supported in configuration)
MCS table	9 entries, Table 1 (custom research table)
Coding rates	1/2, 2/3, 3/4 by puncturing of the (133,171) K = 7 mother code
CRC	CRC-16-CCITT, $x^{16} + x^{12} + x^5 + 1$
EESM β per MCS 0–8	1.5, 1.6, 1.7, 4.5, 5.5, 6.5, 12.0, 16.0, 20.0
OLLA steps	$\Delta_{\text{down}} = 0.5$ dB, $\Delta_{\text{up}} = \Delta_{\text{down}} \cdot p/(1-p)$, offset clamp ± 10 dB
Initial MCS	0
Adaptive stopping	minimum 100 block errors or maximum 2000 blocks per calibration point
Calibration curves	one file per mapping and array (eesm, mean, minimum)
Random seed	7 (master), deterministic per-slot seeding

The first column of Table 4 names each configured aspect and the second column its value; every entry is set in `create_cqi_amc_config`, and the run-configuration files written alongside the results record the values actually used by each executed run.

12. Results and Technical Analysis

Section 12 reports the executed results. Every number of this section is transcribed from the stored CSV files of the seed-7 execution, every figure is generated by the executed scripts, and no value is estimated, extrapolated, or invented. All figures report the method, array size, target BLER, CQI delay, and block counts in their captions; calibration statements quote the block and error counts of the stored files, whose rows also record 95 percent Wilson confidence intervals.

12.1 MCS-specific BLER calibration

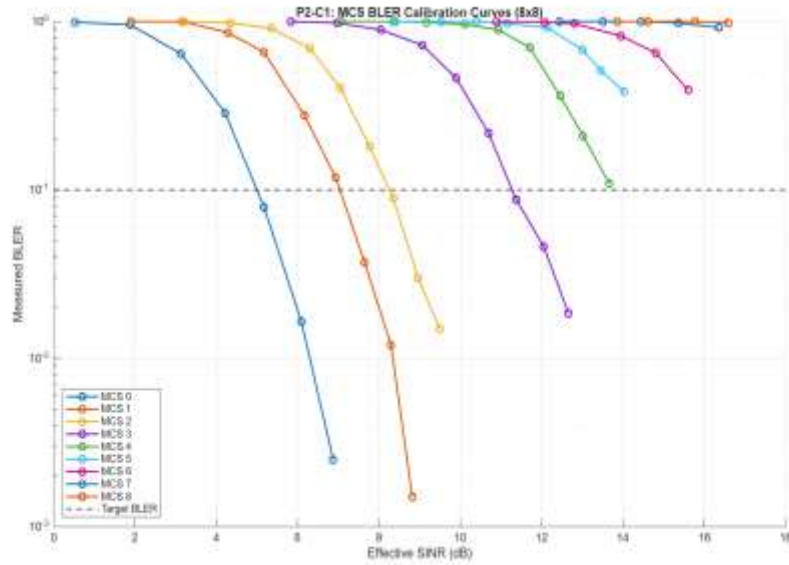


Figure 2. Measured BLER versus effective SINR for MCS 0–8, 8×8 array, adaptive stopping (minimum 100 block errors or 2000 blocks per point), with the 0.10 target line. Per-point block counts range from 104 to 2000.

Figure 2 presents the calibrated BLER curves of the 8×8 array. It shows the nine calibrated curves of the 8×8 array. The measured points preserve strict ordering by MCS index at every overlapping effective SINR, and the spacing between adjacent waterfalls mirrors the spectral-efficiency steps of Table 1. MCS 0 through MCS 3 cross the 0.10 target inside the calibrated windows, each crossing bracketed by the stored points: 4.22 dB effective SINR at BLER 0.2869 and 5.17 dB at 0.0791 for MCS 0, 6.93 dB at 0.1191 and 7.65 dB at 0.0370 for MCS 1, 7.77 dB at 0.1812 and 8.35 dB at 0.0893 for MCS 2, and 10.70 dB at 0.2177 and 11.38 dB at 0.0874 for MCS 3. MCS 4 is not admitted on this array: its lowest stored point is 0.1087 at 13.67 dB effective SINR, carried by 920 blocks and 100 errors, and its 95 percent Wilson interval includes the 0.10 target, so the point estimate leaves MCS 4 above the target and the prediction clamps to that value. MCS 5 through MCS 8 end their windows at 0.3826, 0.3906, 0.9196, and 0.9808 and remain unusable at this target.

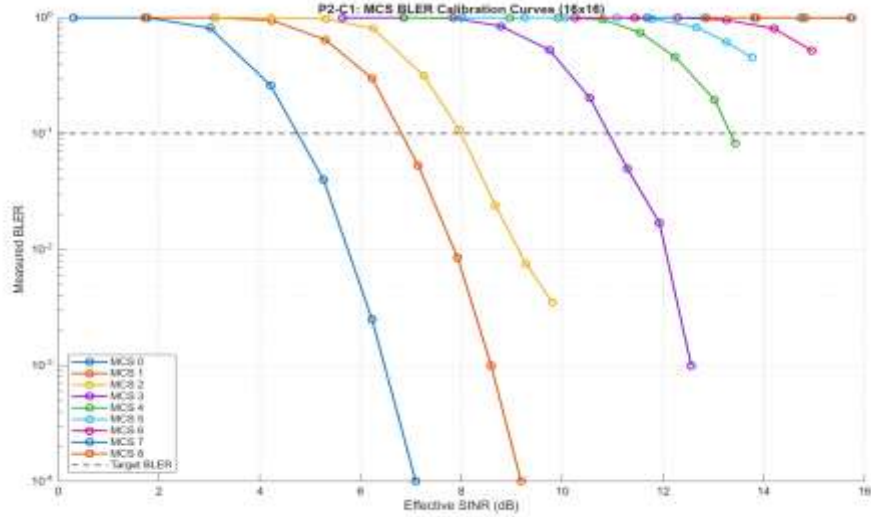


Figure 3. Measured BLER versus effective SINR for MCS 0–8, 16×16 array, same stopping rule and target line. Per-point block counts range from 112 to 2000.

Figure 3 shows the nine calibrated curves of the 16×16 array. The ordering repeats and the waterfalls are steeper than their 8×8 counterparts: MCS 0 falls from 0.2604 to 0.0400 between 4.22 and 5.27 dB effective SINR and reaches a measured zero in 2000 blocks by 7.10 dB. The decisive difference from the 8×8 calibration is MCS 4, which crosses the target on this array, passing from 0.1953 at 13.03 dB to a below-target 0.0822 at 13.43 dB carried by 1216 blocks and 100 errors, so the 16×16 selector gains a fifth admissible entry. The remaining crossings are bracketed at 7.15 dB at 0.0530 for MCS 1, at 7.96 dB at 0.1078 and 8.68 dB at 0.0240 for MCS 2, and at 10.55 dB at 0.2036 and 11.29 dB at 0.0500 for MCS 3, while MCS 5 through MCS 8 end their windows at 0.4554, 0.5208, 0.9911, and 1.0000.

Figure 4 presents the configured EESM calibration parameter against the MCS index.

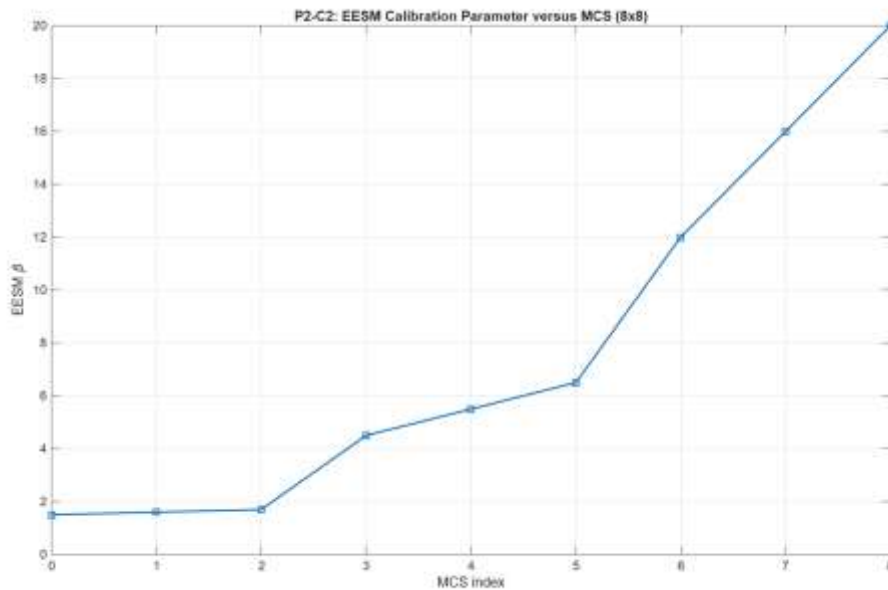


Figure 4. Configured EESM calibration parameter β versus MCS index.

Figure 4 shows the configured β values of Table 4 against the MCS index. The profile is monotone and three-tiered: the QPSK entries use 1.5, 1.6, and 1.7, the 16-QAM entries step to 4.5, 5.5, and 6.5, and the 64-QAM entries climb to 12.0, 16.0, and 20.0, which places the mapping near the minimum for the fragile low-order codewords and closer to the mean for the high-order codewords.

Figure 5 presents the leave-one-out self-consistency of the 8×8 calibration.

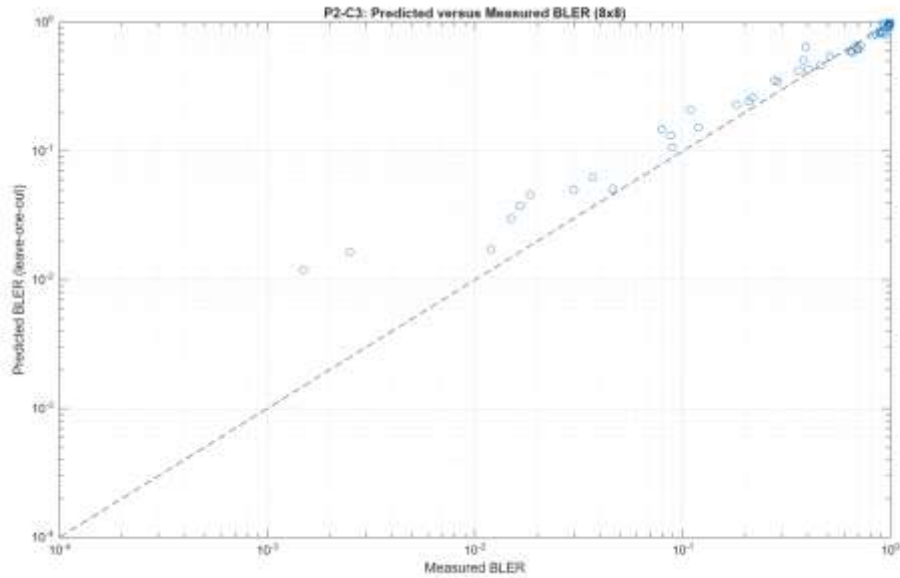


Figure 5. Leave-one-out predicted versus measured BLER across all calibration points, 8×8 array.

Figure 5 shows the self-consistency of the calibrated prediction on the 8×8 array. The leave-one-out points concentrate along the identity line at measured BLER above roughly 0.1, and the prediction turns conservative below that region, lying above the identity line at the lowest measured error rates where the finite block budget limits resolution. A conservative low-BLER bias can only delay, never hasten, the admission of a marginal MCS.

Figure 6 presents the corresponding self-consistency of the 16×16 calibration.

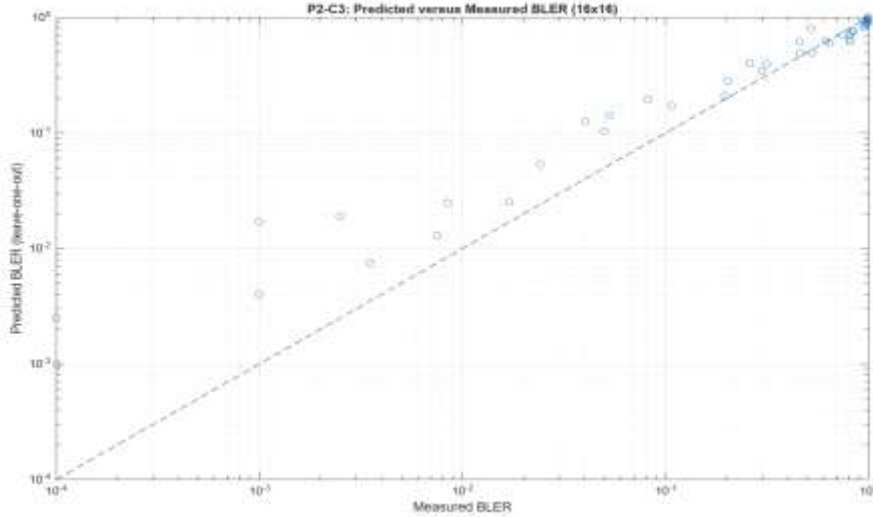


Figure 6. Leave-one-out predicted versus measured BLER across all calibration points, 16×16 array.

Figure 6 shows the corresponding self-consistency on the 16×16 array. The scatter repeats the 8×8 character, with tight agreement along the identity line through the decision region and a conservative upward bias at the smallest measured BLER values. No point falls materially below the identity line around the target, so the prediction does not admit an MCS that the measurement would reject.

12.2 Effective-SINR mapping comparison

Section 12.2 compares the mean-SINR controller with the EESM controller on identical realizations, each querying the curves calibrated for its own mapping. With mapping-matched curves the mean controller no longer reads its curves optimistically: on the 8×8 array its measured BLER stays at or below the target at every point from the 12 dB crossing onward, with a worst compliant value of 0.073, and it delivers 52540 bits per slot at 30 dB against 52341 for the EESM controller. On the 16×16 array the mean controller is also compliant after the crossing, with a worst compliant value of 0.046, and delivers 112297 bits per slot at 30 dB against 106457 for the EESM controller. The mean controller therefore reaches a slightly higher terminal goodput because it climbs to a higher average CQI, and the two mappings are now a controlled comparison rather than the miscalibration artifact of the earlier build, in which the mean controller queried EESM-calibrated curves. The difference between the two mappings at this target is small, and both remain inside the reliability constraint across the compliant region.

12.3 CQI accuracy

Section 12.3 reports the average CQI of every method. The agreement requirement of Section 4 is answered by the overlap of three curves: the delayed EESM controller, the ideal same-slot variant, and the pilot-time-CSI reference are indistinguishable across the sweep on both arrays, so the receiver-estimated CQI reproduces the perfect pilot-time-CSI CQI on average at every operating point. The mean-SINR curve climbs a little earlier than the EESM family, the OLLA curve leads the EESM family slightly because its positive offset raises the effective decision variable, the fixed-MCS reference remains at zero by construction, and the threshold rule reports the staircase of its configured SNR thresholds rather than any receiver quantity.

Figure 7 presents the reported CQI of every method on the 8×8 array.

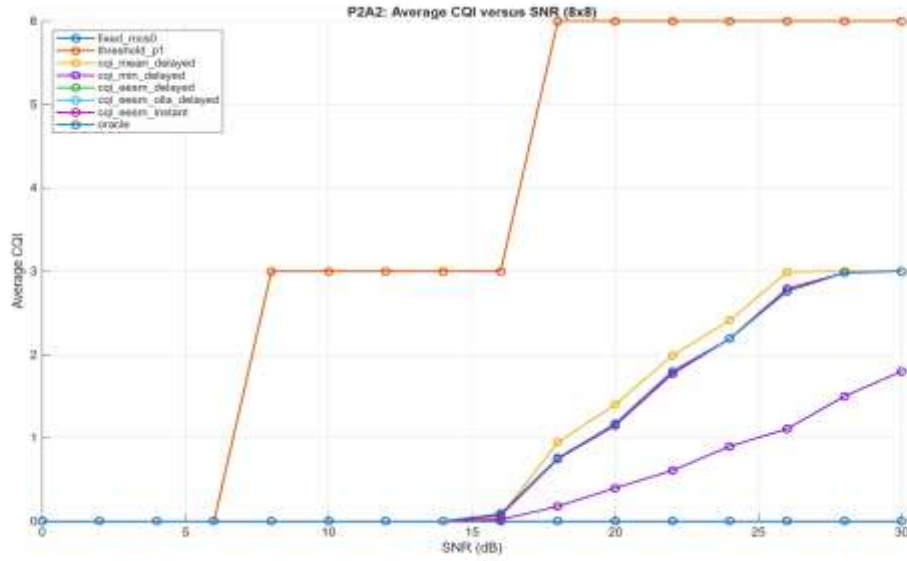


Figure 7. Average CQI versus SNR, all methods, 8×8 array, target BLER 0.10, CQI delay 1 slot, 100 slots per point.

Figure 7 shows the reported CQI of every method across the sweep on the 8×8 array. The EESM triplet rises together from 16 dB, passes an average CQI of two near 22 dB, and saturates at three, the highest admissible entry of the 8×8 calibration. The threshold staircase jumps from zero to three at 8 dB and from three to six at 18 dB, values chosen by the configured SNR alone.

Figure 8 presents the corresponding CQI behavior on the 16×16 array.

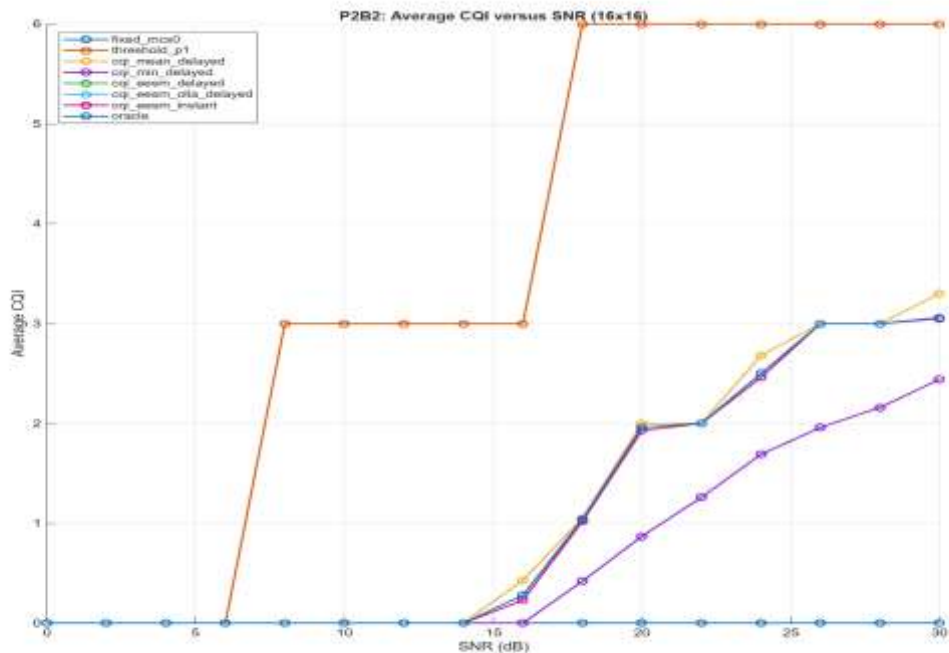


Figure 8. Average CQI versus SNR, all methods, 16×16 array, same configuration.

Figure 8 shows the corresponding CQI behavior on the 16×16 array. The EESM triplet again moves as one curve, reaching an average of two at 20 dB and about 3.05 at 30 dB once the

Figure 10 presents the corresponding compliance picture on the 16×16 array.

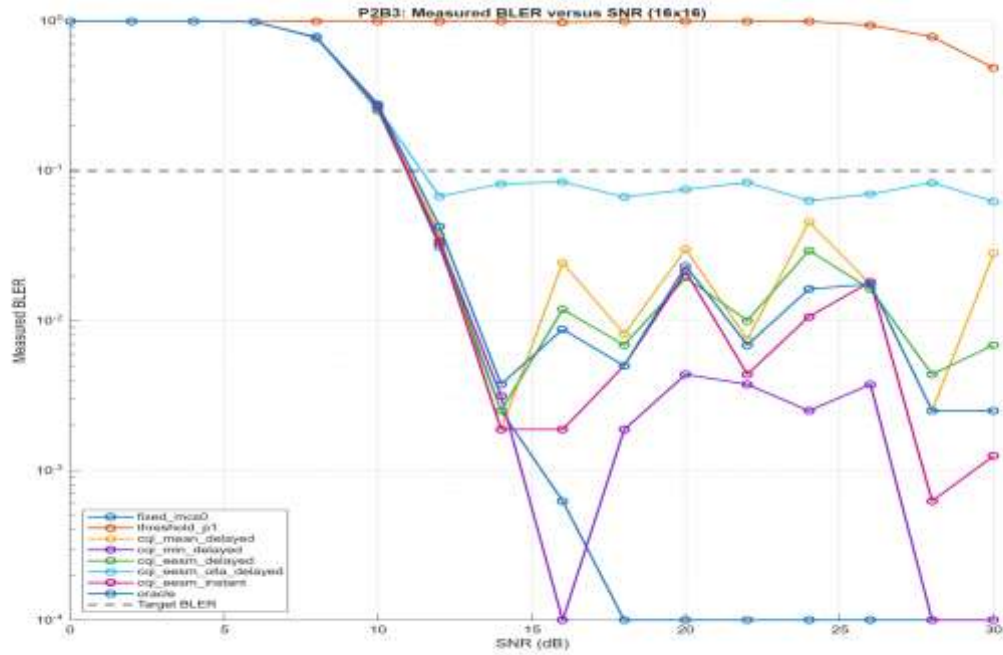


Figure 10. Measured BLER versus SNR with the target line, all methods, 16×16 array (1600 blocks per method per point).

Figure 10 shows the corresponding compliance picture on the 16×16 array. The primary controller crosses at 12 dB with 0.034 and stays compliant through 30 dB, with stored values of 0.034, 0.003, 0.012, 0.007, 0.019, 0.010, 0.029, 0.016, 0.004, and 0.007 and an average of 0.014 over the compliant region, deeper below the target than the 8×8 run. The threshold rule again pins measured BLER near one through most of the sweep and remains at 0.486 at 30 dB, the mean-SINR method is compliant after its crossing, and the OLLA trace rides just under the target line across the compliant region.

Figure 11 presents the average selected MCS of every method on the 8×8 array.

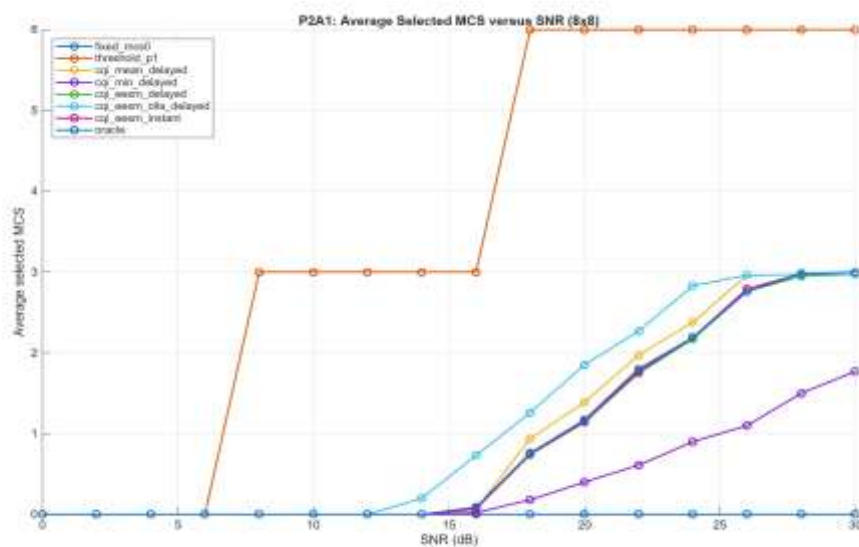


Figure 11. Average selected MCS versus SNR, all methods, 8×8 array.

Figure 11 shows how each controller climbs the MCS table across the sweep. The three EESM curves are indistinguishable and saturate at an average MCS of about three, the admissibility ceiling of the 8×8 calibration. The threshold staircase leaps to six at 18 dB, a selection three steps above anything the calibrated constraint admits, which is the selection error that Figure 9 prices in block errors.

Figure 12 presents the corresponding MCS climb on the 16×16 array.

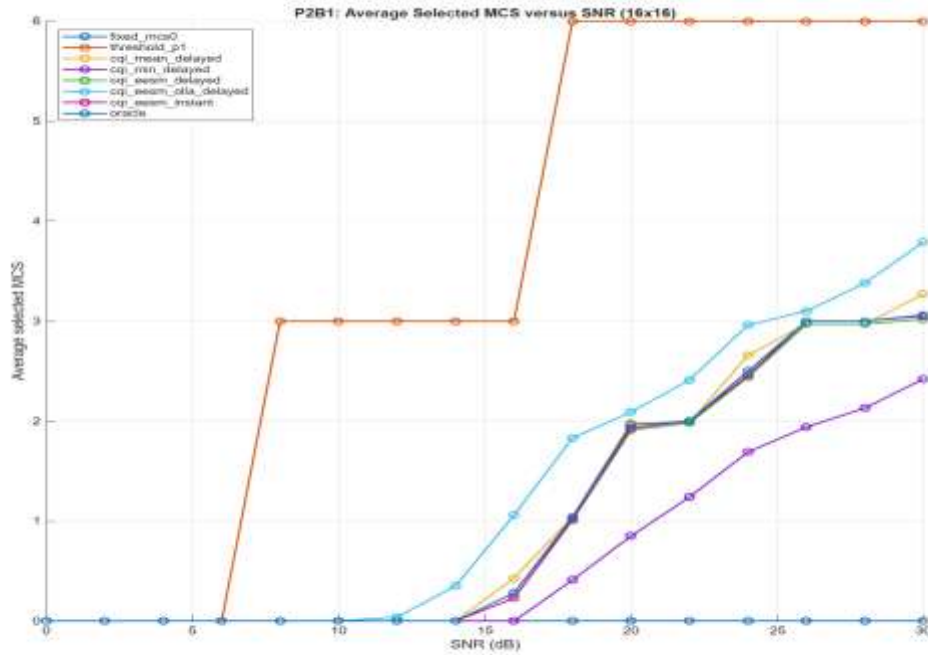


Figure 12. Average selected MCS versus SNR, all methods, 16×16 array.

Figure 12 shows the corresponding MCS climb on the 16×16 array. The EESM triplet passes an average of two at 20 dB, pauses at three, and rises to about 3.02 at 30 dB as the newly admissible MCS 4 takes a small selection share, and the threshold and fixed references repeat their array-independent behavior.

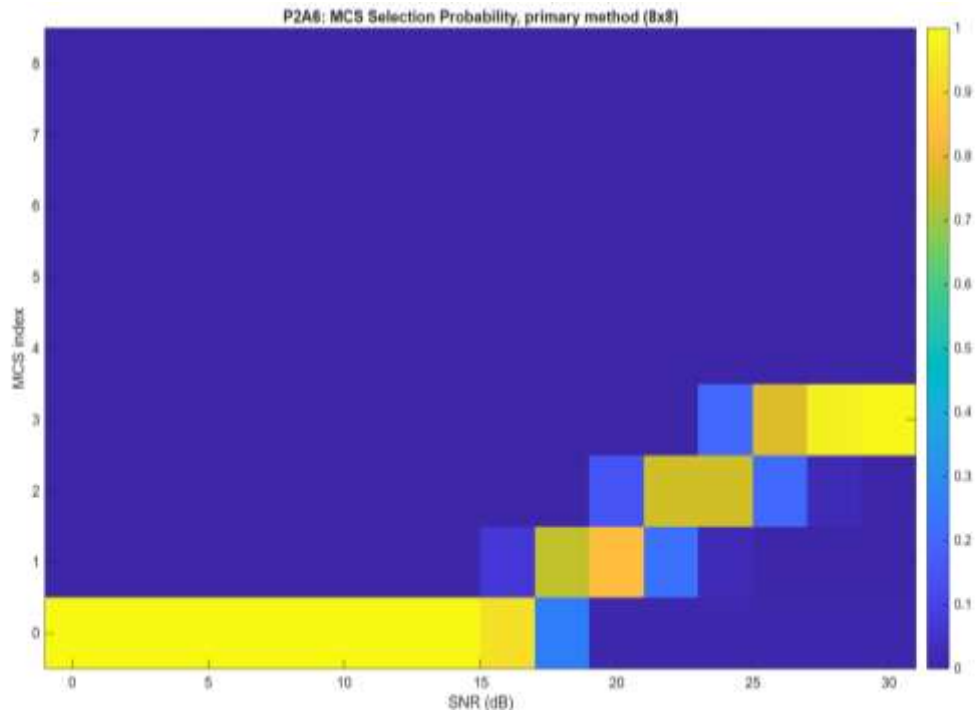


Figure 13. MCS selection probability versus SNR, primary method (delayed EESM), 8×8 array.

Figure 13 presents the selection distribution of the primary method on the 8×8 array. It shows the selection distribution of the primary method. Below 15 dB the selection sits on MCS 0, the staircase then hands over through MCS 1 and MCS 2, and from 26 dB the mass concentrates on MCS 3, the 8×8 ceiling. No mass appears above MCS 3 at any SNR, the distributional form of the compliance result: the selector never gambles on an entry whose calibrated prediction misses the target. Over the whole sweep the primary method spends 57.8 percent of slots on MCS 0, 11.9 percent each on MCS 1 and MCS 2, and 18.4 percent on MCS 3, with no selections of MCS 4 or above.

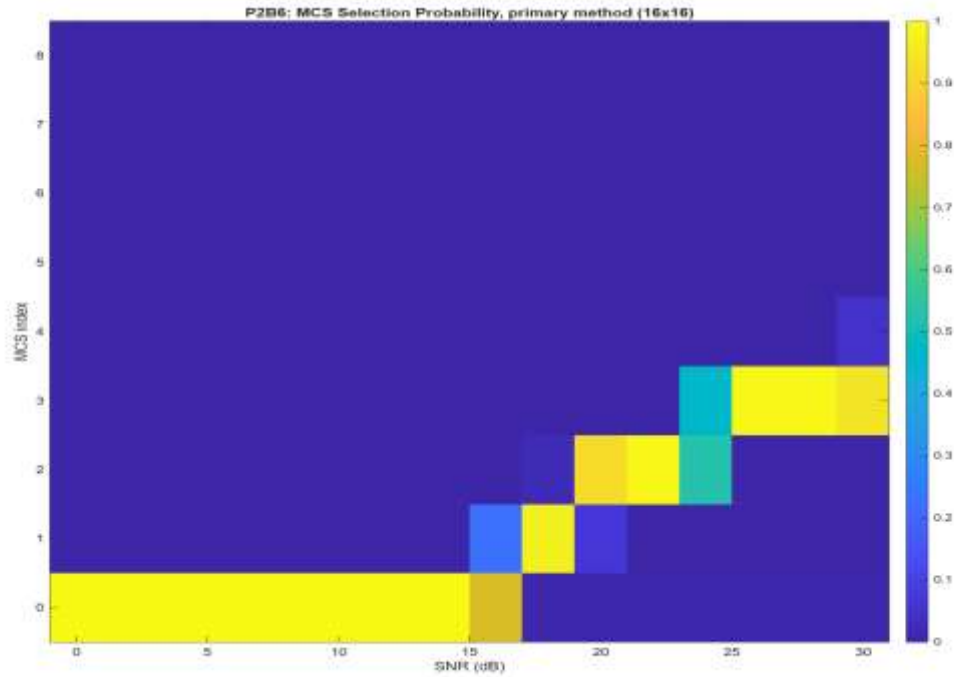


Figure 14. MCS selection probability versus SNR, primary method, 16×16 array.

Figure 14 shows the corresponding distribution at 16×16. The staircase is sharper than the 8×8 one, with near-unit probability plateaus on MCS 2 and MCS 3, and at 30 dB a thin share of MCS 4 appears. Over the whole sweep the primary method spends 55.3 percent of slots on MCS 0, 7.9 percent on MCS 1, 15.4 percent on MCS 2, 21.1 percent on MCS 3, and 0.3 percent on MCS 4, the direct fingerprint of the MCS 4 admissibility that the 16×16 calibration granted.

Figure 15 presents the per-slot switching activity of every controller on the 8×8 array.

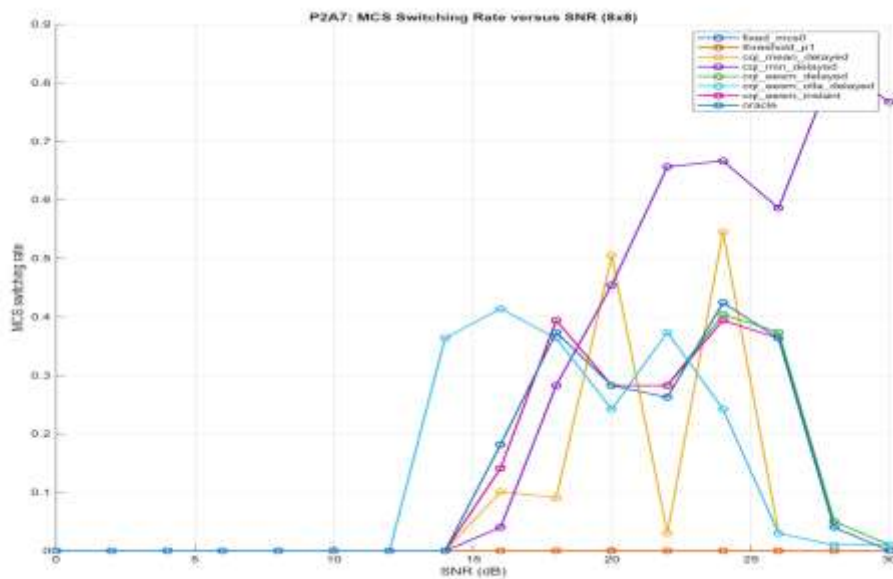


Figure 15. MCS switching rate versus SNR, all methods, 8×8 array.

Figure 15 shows the per-slot switching activity of every controller. The fixed and threshold references never switch, and the receiver-driven methods peak where their selection straddles a table boundary and fall toward zero at 30 dB where the selection settles on the ceiling entry. The minimum-mapping controller sustains the highest switching because its worst-element statistic hunts across boundaries.

Figure 16 presents the corresponding switching activity on the 16×16 array.

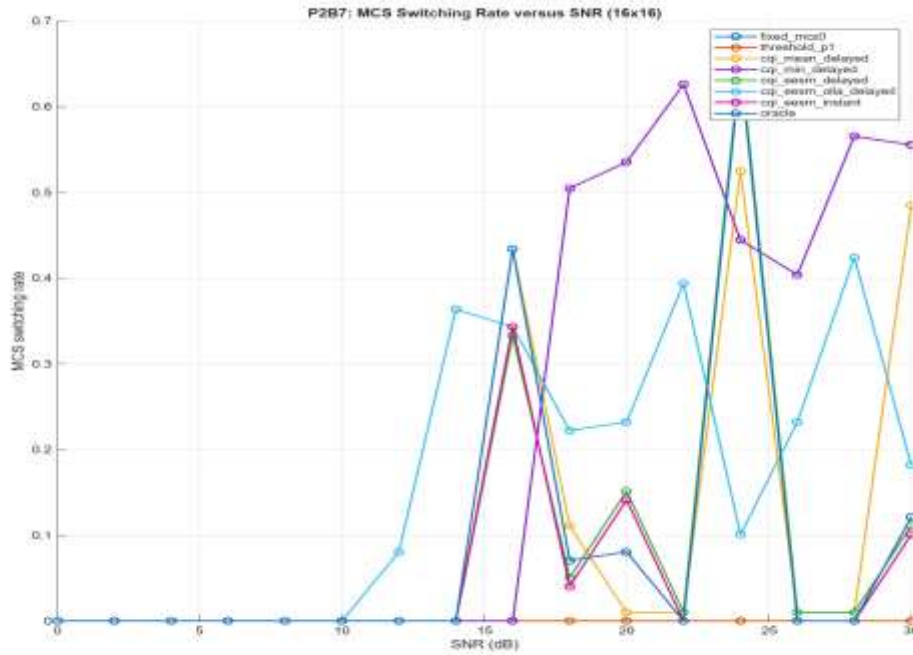


Figure 16. MCS switching rate versus SNR, all methods, 16×16 array.

Figure 16 shows the corresponding switching activity at 16×16. The receiver-driven methods peak in the transition bands, the minimum-mapping controller again shows the widest and highest switching, and the primary method drops toward zero where the effective SINR sits squarely inside one MCS region.

12.5 Goodput and spectral efficiency

Figure 17 presents the delivered information per slot of every method on the 8×8 array.

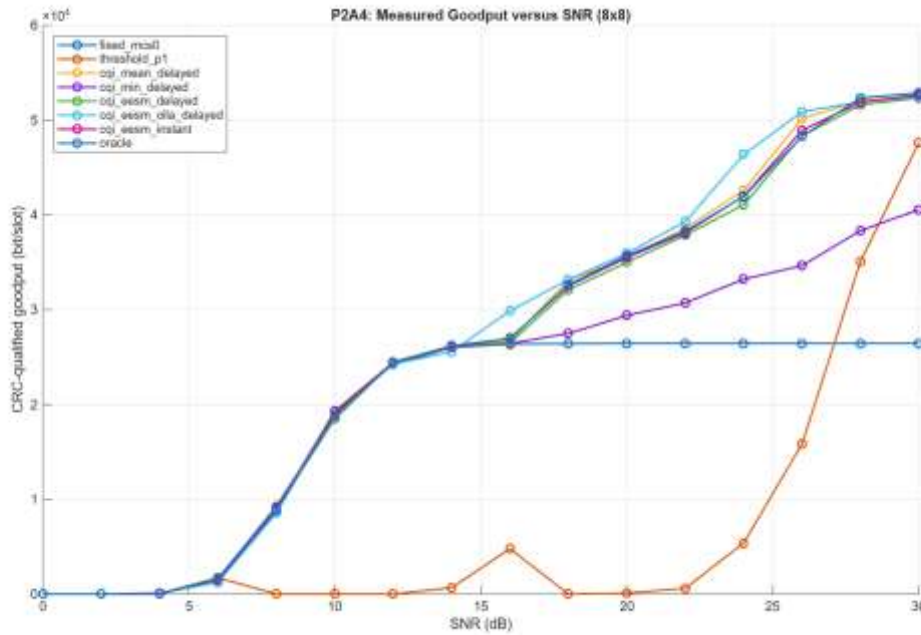


Figure 17. CRC-qualified goodput versus SNR, all methods, 8×8 array.

Figure 17 shows the delivered information per slot on the 8×8 array. All receiver-driven methods ride one curve up to about 14 dB, where the fixed reference saturates at its rate-half QPSK ceiling of 26448 bits per slot, and the adaptive methods then separate and climb to 52341 bits per slot for the primary controller at 30 dB, a factor of 1.98 above the fixed ceiling. The threshold baseline delivers almost nothing between 8 and 24 dB because nearly every block fails its CRC, recovers only at the top of the sweep, and ends at 47618 bits per slot, 9.9 percent below the primary controller at the single SNR most favorable to it. The minimum-mapping controller ends lowest of the adaptive methods at 40558 bits per slot.

Figure 18 presents the corresponding goodput on the 16×16 array.

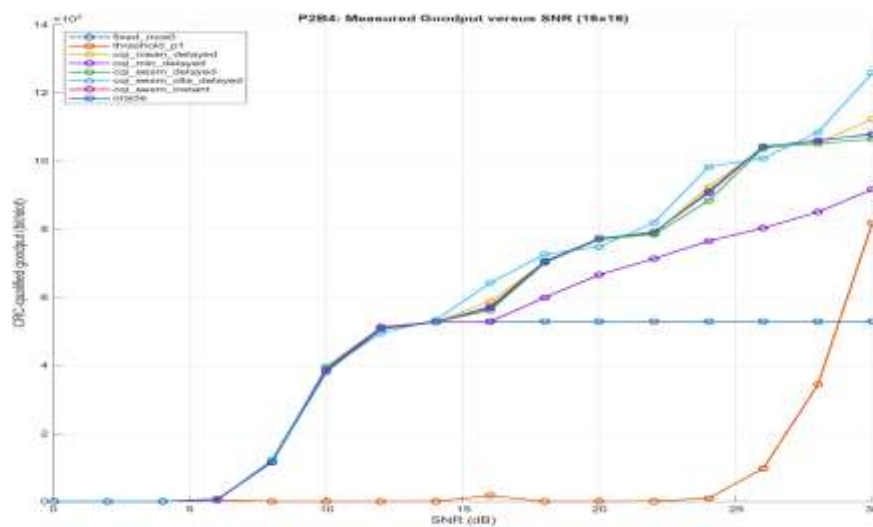


Figure 18. CRC-qualified goodput versus SNR, all methods, 16×16 array.

Figure 18 shows the corresponding goodput at 16×16. The fixed ceiling doubles to 52896 bits per slot, the primary controller reaches 106457 bits per slot at 30 dB, a factor of 2.01 above that

ceiling, and the practical, instant, and reference curves remain within a few percent of one another. The OLLA curve ends highest at 125860 bits per slot because it operates closer to the target, the mean-mapping curve ends at 112297, and the threshold baseline stays near zero until 24 dB and ends at 81888 bits per slot, 30.0 percent below the primary controller.

Figure 19 presents the per-resource-element view of the delivered payload on the 8×8 array.

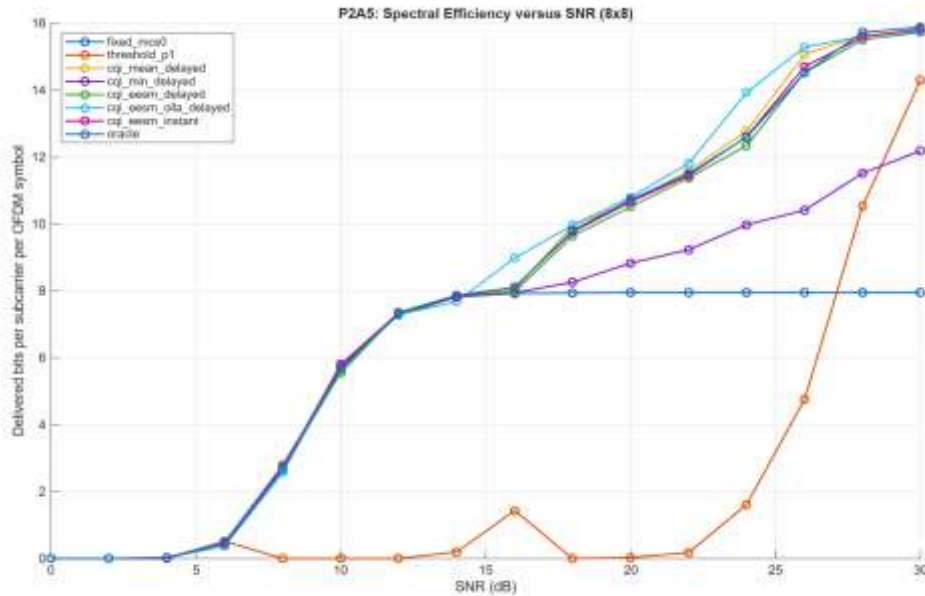


Figure 19. Delivered spectral efficiency versus SNR, all methods, 8×8 array.

Figure 19 shows the per-resource-element view of the delivered payload. The normalization removes the slot geometry and leaves the ordering of Figure 17 unchanged: the fixed reference plateaus at just under eight delivered bits per subcarrier per OFDM symbol across the eight layers, and the primary controller ends just under sixteen.

Figure 20 presents the corresponding view on the 16×16 array.

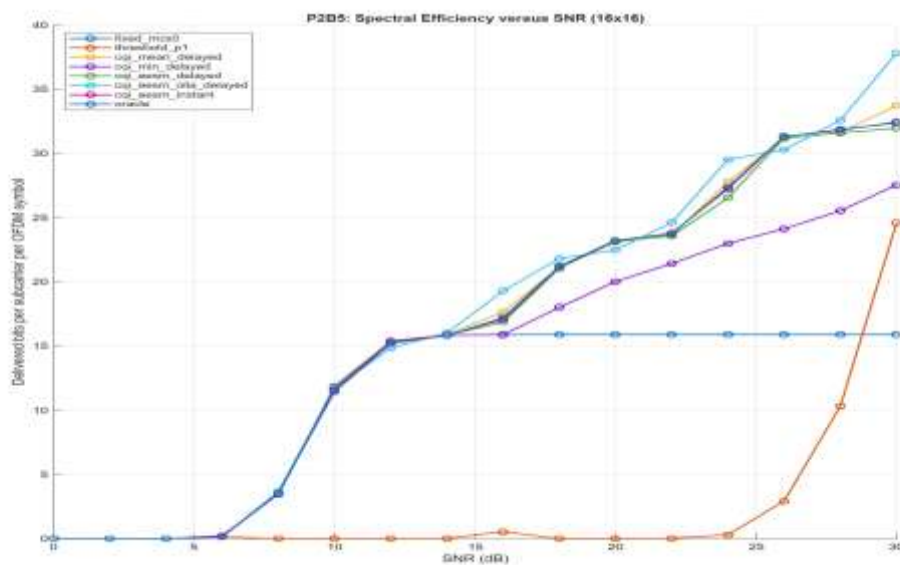


Figure 20. Delivered spectral efficiency versus SNR, all methods, 16×16 array.

Figure 20 shows the corresponding view at 16×16 . The fixed plateau sits just under sixteen delivered bits per subcarrier per OFDM symbol across the sixteen layers, and the OLLA curve ends highest, so the spectral-efficiency ordering is the goodput ordering rescaled and no additional mechanism appears at the per-resource-element level.

12.6 Feedback-delay impact

Section 12.6 reads the one-slot delay from the stored files. On the 8×8 array `cqi_eesm_instant` ends at 52674 bits per slot against 52341 for the delayed controller, a difference of 333 bits per slot, and the two BLER curves of Figure 9 overlap within the resolution of the plot. On the 16×16 array the instant variant ends at 107786 bits per slot against 106457, a gap of 1.25 percent concentrated at the top of the sweep, where the selection alternates between MCS 3 and MCS 4 and a one-slot-old report occasionally applies the lower entry a slot too long. The same-slot variant is presented strictly as the causality-free upper bound, so the deployable controller forfeits between a fraction of a percent and about one and a quarter percent of the bound.

Figure 21 presents the one-slot lag between the report and its application on the 8×8 array.

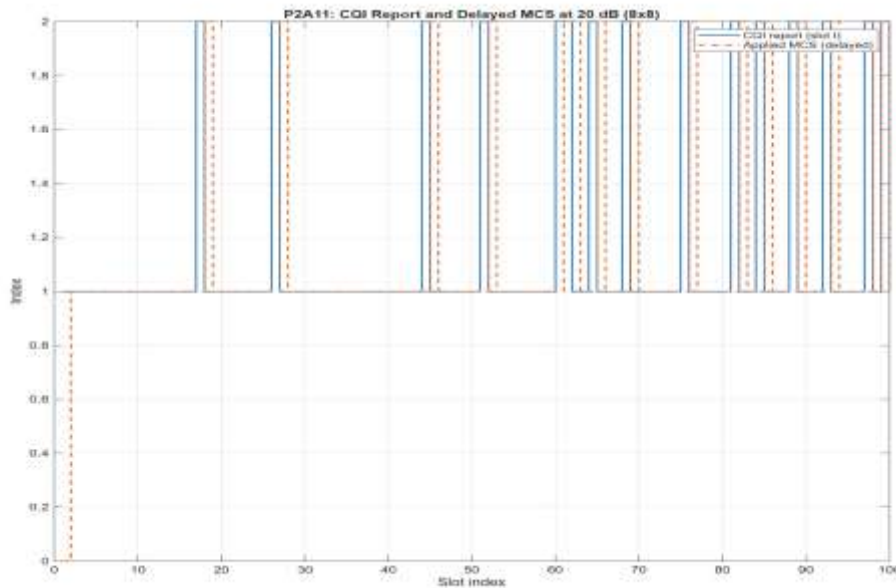


Figure 21. CQI report and delayed applied MCS versus slot index at the convergence SNR, primary method, 8×8 array.

Figure 21 shows the one-slot lag between the report and its application. The applied trace is the report trace shifted right by exactly one slot, with slot one driven by the configured initial MCS 0 before the first report arrives, which is the causality that Gate 10 verifies. At this operating point the report alternates between adjacent entries, so the delayed application regularly transmits one entry for one slot after the report has already moved.

12.7 OLLA convergence

Figure 22 presents the outer-loop offset trajectory on the 8×8 array.

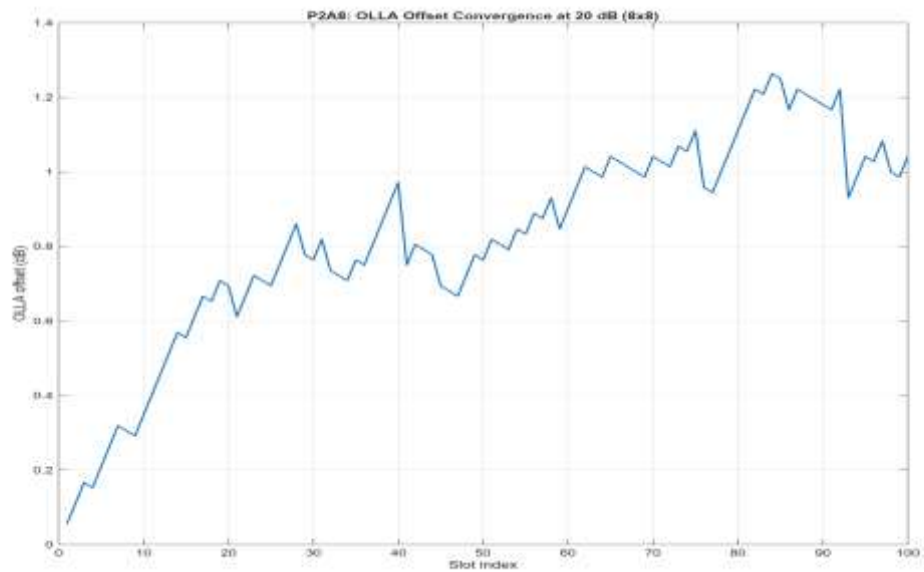


Figure 22. OLLA offset versus slot index at the convergence SNR, 8×8 array.

Figure 22 shows the outer-loop offset trajectory. From its zero initialization the offset rises in the failure-corrected sawtooth of the asymmetric update, each block failure stepping it down and each success raising it by the target-locked fraction, and over the logged window it settles around 0.8 dB without diverging or railing against the clamp. The positive offset means the corrected loop operates the link slightly more aggressively than the calibrated prediction alone, which is the mechanism that lifts the running BLER toward the target rather than below it.

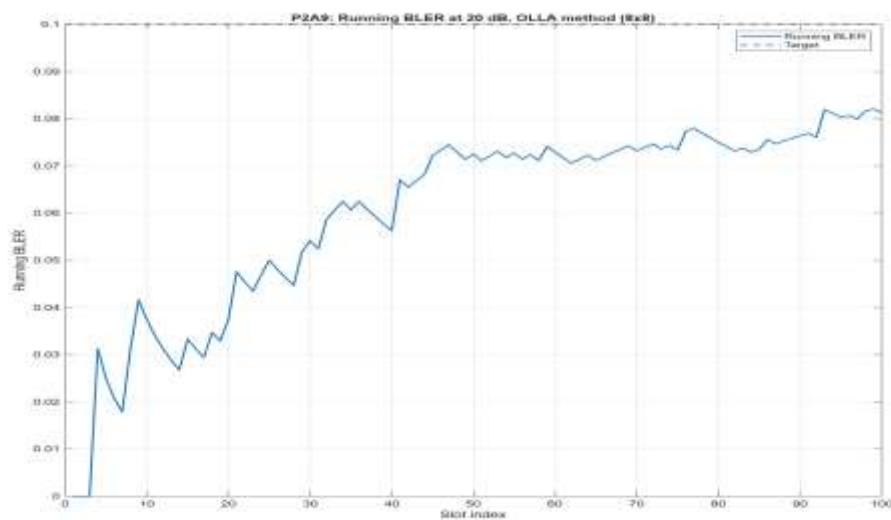


Figure 23. Running BLER versus slot index at the convergence SNR with the target line, OLLA method, 8×8 array.

Figure 23 shows the running block error rate of the outer-loop method against the target. After an initial transient the running rate climbs and settles just below the 0.10 line, which is the intended operating point of an outer loop that updates on the block verdicts and drives the block

error rate toward the target. Across the compliant region the OLLA method holds a measured BLER between 0.006 and 0.088 and delivers the highest terminal goodput of any method on this array.

Figure 24 presents the per-slot MCS decisions of the primary method on the 8×8 array.

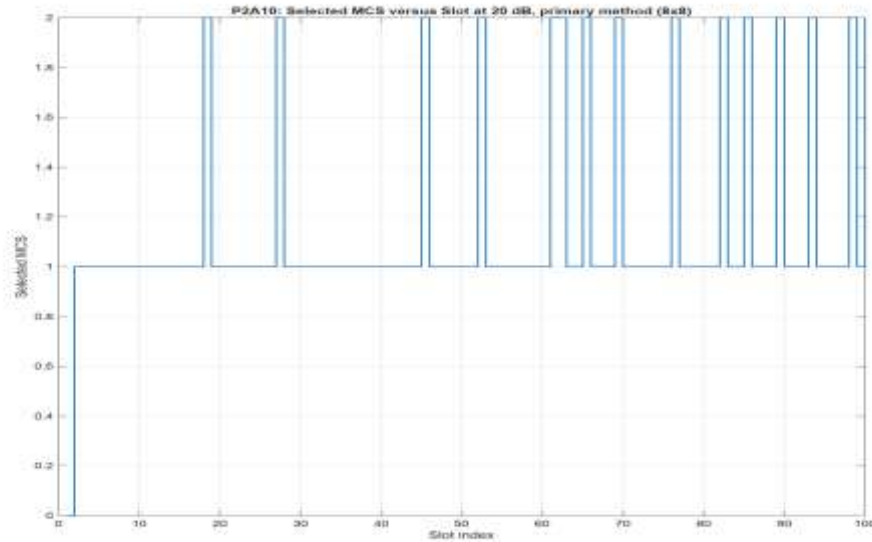


Figure 24. Selected MCS versus slot index at the convergence SNR, primary method, 8×8 array.

Figure 24 shows the per-slot MCS decisions of the primary method. At this operating point the controller alternates between adjacent entries for the whole logged window after the initial MCS 0 slot, so the switching rate of Figure 15 at this SNR is generated by first-neighbor alternation across one table boundary rather than by hunting over the table.

Figure 25 presents the outer-loop offset trajectory on the 16×16 array.

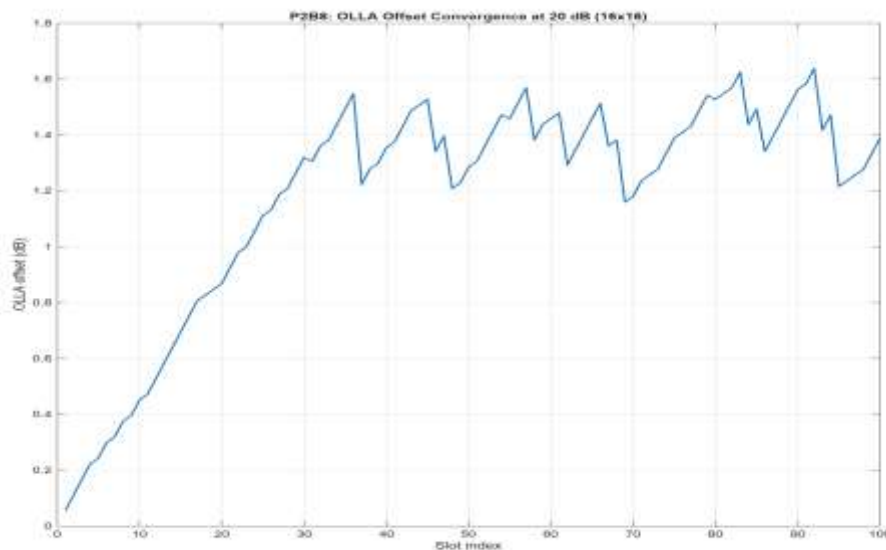


Figure 25. OLLA offset versus slot index at the convergence SNR, 16×16 array.

Figure 25 shows the outer-loop offset trajectory on the 16×16 array. The sawtooth repeats the 8×8 character and settles higher, around 1.2 dB, so the larger array sustains a larger positive

offset while remaining stable and clamp-free, consistent with its steeper calibration and larger per-slot block count.

Figure 26 presents the corresponding running block error rate on the 16×16 array.

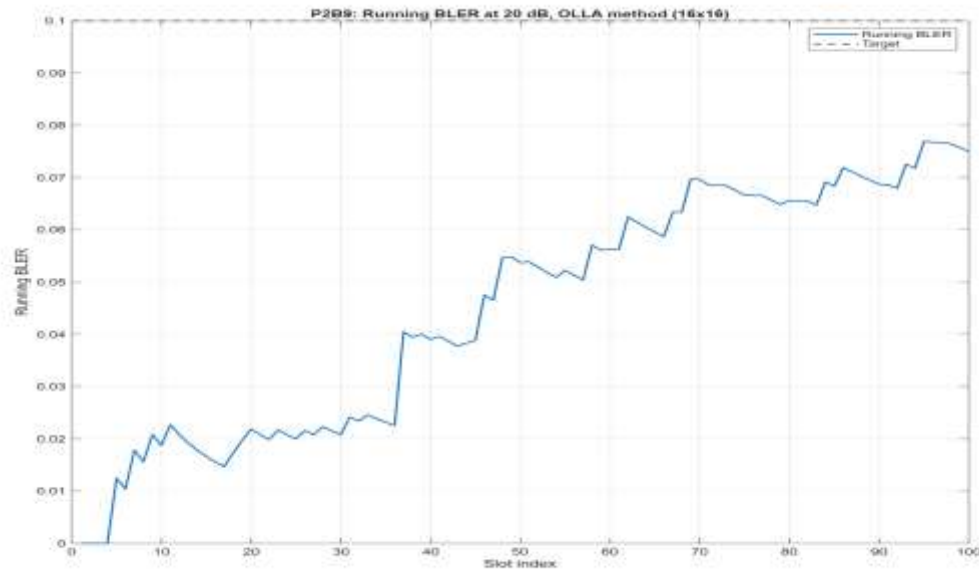


Figure 26. Running BLER versus slot index at the convergence SNR with the target line, OLLA method, 16×16 array.

Figure 26 shows the corresponding running block error rate. The transient settles just under the 0.10 target and holds there for the remainder of the window, the same target-tracking behavior as the 8×8 loop, and across the compliant region the 16×16 OLLA method holds a measured BLER between 0.063 and 0.084.

Figure 27 presents the per-slot MCS decisions of the primary method on the 16×16 array.

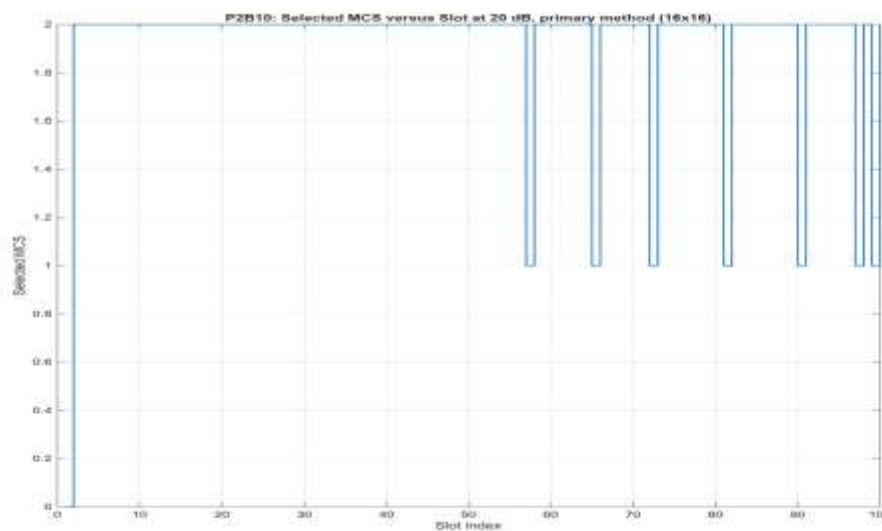


Figure 27. Selected MCS versus slot index at the convergence SNR, primary method, 16×16 array.

Figure 27 shows the per-slot MCS decisions of the primary method on the 16×16 array. After the initial MCS 0 slot the selection sits close to constant for the logged window, the trace behind the low switching rate of Figure 16 at this SNR, because the 16×16 effective SINR at this operating point lies inside one MCS region rather than straddling a boundary.

Figure 28 presents the report and its delayed application on the 16×16 array.

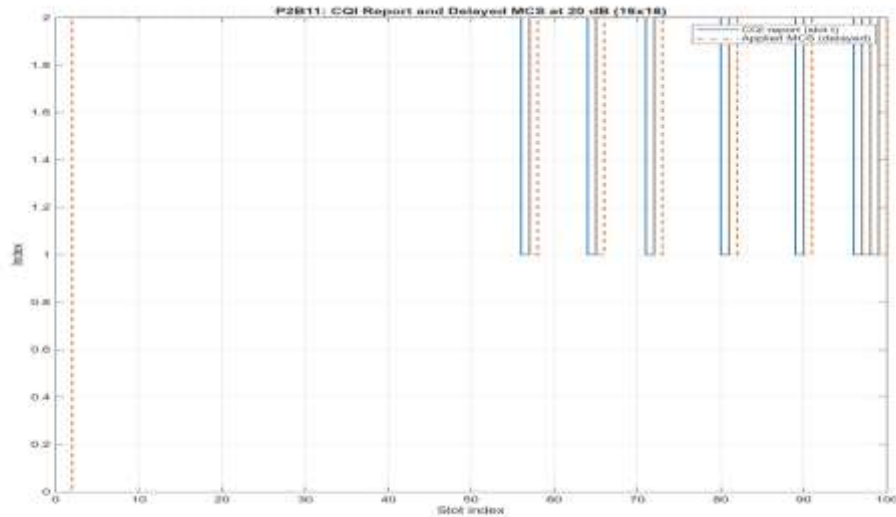


Figure 28. CQI report and delayed applied MCS versus slot index at the convergence SNR, primary method, 16×16 array.

Figure 28 shows the report and its delayed application. The applied trace equals the report trace shifted right by one slot, and because the report is close to constant at this operating point the two traces coincide except at the first applied slot, which carries the configured initial MCS 0, so the causality of Gate 10 is visible even in a near-constant-selection regime.

12.8 8×8 versus 16×16 comparison

Figure 29 presents the cross-array compliance comparison.

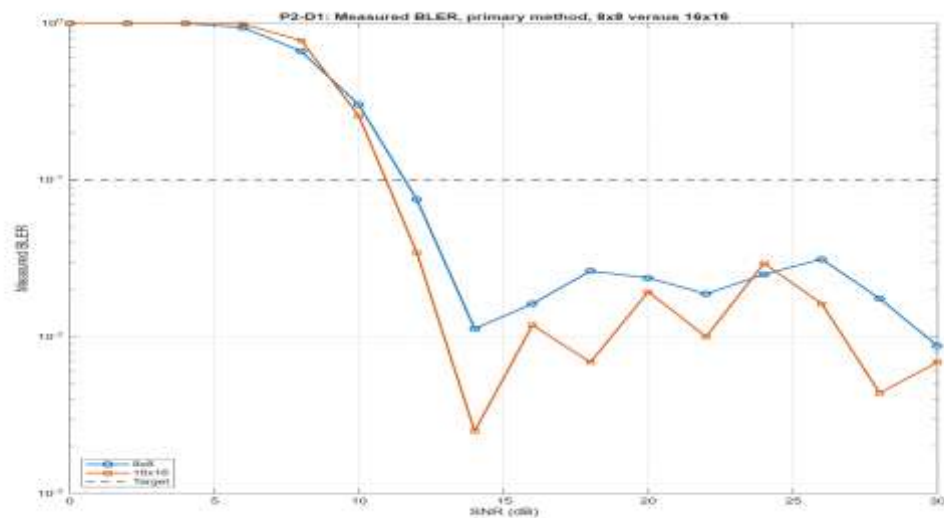


Figure 29. Measured BLER of the primary method, 8×8 versus 16×16, with the target line.

Figure 29 shows the cross-array compliance comparison. Both curves fall together through the waterfall, both reach the target at 12 dB, with measured values of 0.075 at 8×8 and 0.034 at 16×16 at the crossing, and both remain below the target at every subsequent sweep point, the 8×8 curve averaging 0.025 and the 16×16 curve 0.014 over the compliant region. Neither curve exhibits a single post-crossing violation, so the compliance property of the controller is array-independent across the doubling of the spatial dimension.

Figure 30 presents the cross-array goodput comparison.

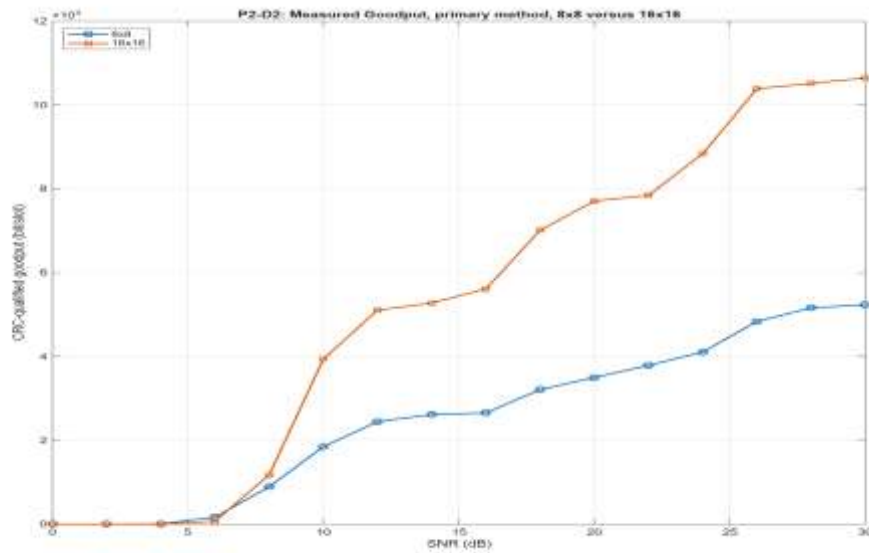


Figure 30. CRC-qualified goodput of the primary method, 8×8 versus 16×16 .

Figure 30 shows the cross-array goodput comparison. The two curves separate at 8 dB and the 16×16 curve runs approximately twice the 8×8 curve across the upper half of the sweep, ending at 106457 against 52341 bits per slot at 30 dB, a measured factor of 2.03 that matches the doubling of the spatial streams. The fixed-MCS references confirm the arithmetic, doubling exactly from 26448 to 52896 bits per slot.

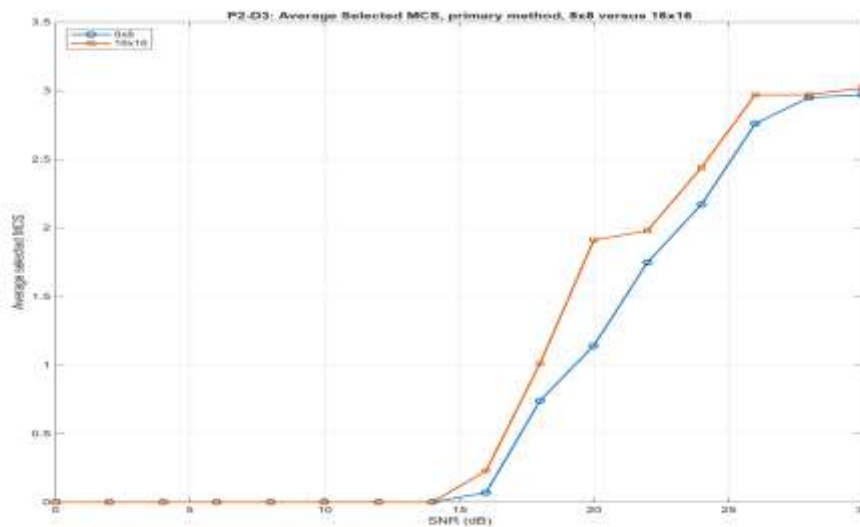


Figure 31. Average selected MCS of the primary method, 8×8 versus 16×16 .

Figure 31 shows the cross-array selection comparison. The 16×16 controller climbs slightly earlier through the transition band, matches the 8×8 controller at an average MCS of two near 20 to 22 dB, and separates above 28 dB toward an average near 3.02 while the 8×8 controller saturates just below three. The larger array admits MCS 4 within the calibrated range in a small fraction of slots, while the 8×8 array does not; because the executed quality scalar is now formed per layer, the earlier climb reflects the steeper measured calibration of the larger array rather than an aggregation artifact.

12.9 Comparison with the Project 1 threshold adaptation

Figure 32 presents the practical controller against the pilot-time-CSI reference on both arrays.

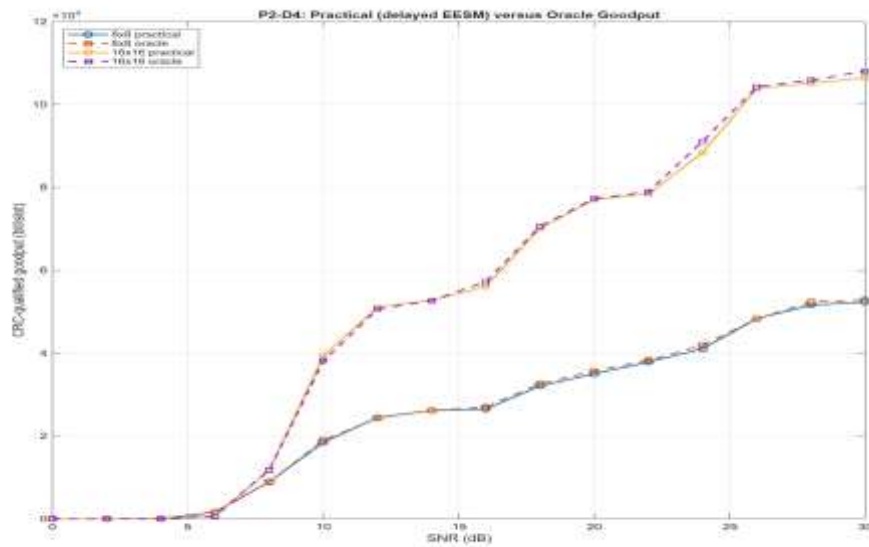


Figure 32. Practical delayed-EESM controller versus the pilot-time-CSI reference, both arrays.

Figure 32 shows the distance between the practical controller and the pilot-time-CSI reference. On the 8×8 array the two curves are close at every sweep point and their terminal values differ by 532 bits per slot, 52341 against 52873, so channel-estimation error and the one-slot delay together cost the smaller array about one percent. On the 16×16 array the curves coincide until about 22 dB and a gap opens over the last quarter of the sweep, ending at 106457 against 108008 bits per slot, a 1.4 percent deficit concentrated where the MCS 3 to MCS 4 boundary makes the selection most sensitive to report staleness and estimation noise. The practical controller therefore operates within about one and a half percent of the reference everywhere; because the reference is perfect only at pilot time and is evaluated on curves calibrated against estimated SINR, it bounds the reporting loss rather than the full link, which is why the outer-loop method can exceed it by operating closer to the target.

The direct confrontation of the study is quantitative on identical realizations. The threshold rule of the baseline study violates the 0.10 target across essentially its entire useful range on both arrays: from 8 dB through the mid-twenties its measured BLER sits near one because its SNR thresholds select modulation orders the post-equalization SINR cannot carry, and even at 30 dB it remains at 0.403 on the 8×8 array and 0.486 on the 16×16 array. Across that same range it delivers almost no CRC-qualified goodput, while the receiver-driven controller delivers tens of thousands of bits per slot in full compliance. At 30 dB, the single point most favorable to the threshold rule, the comparison still favors the CQI controller: 52341 against 47618 bits per slot

on the 8×8 array, a gain of 9.9 percent, and 106457 against 81888 on the 16×16 array, a gain of 30.0 percent. The receiver-driven controller trades no goodput for its BLER compliance, and instead gains goodput across the sweep on both arrays while converting an almost always non-compliant link into an always compliant one.

13. Limitations

Section 13 states the simplifications honestly. The MCS table is a nine-entry custom research table, not a standardized table. The channel code is a punctured convolutional code with CRC-16, not the LDPC coding and CRC-24 segmentation of standardized modern data channels. The channel is the five-tap TDL-A profile with a simplified common-Doppler phase-evolution model rather than a full geometric or spatially correlated model. CQI feedback is simplified: one wideband report per slot, an integer index without quantization modeling beyond the optional reporting-error experiment, and a fixed delay. One wideband MCS serves all layers of a slot; there is no per-layer MCS, no rank adaptation, no HARQ, no scheduler, no multi-user interference, and no full NR slot structure. The EESM prediction depends on its calibration parameters, which are configured values validated by the predicted-versus-measured comparison of the executed run. The SINR estimate is formed once per slot at the pilot time and held constant across the data symbols. Entries 5 and 6 of the MCS table share the same nominal spectral efficiency of 3.00 bits per symbol; neither entry is admitted in the executed sweeps, so the highest-index selection rule has no effect on the reported results, and a spectral-efficiency-first tie-break is available in configuration. MCS 4 is admitted only on the 16×16 array and only in a small fraction of slots at the top of the sweep, and its 8×8 admissibility is left statistically open by the Wilson interval of its lowest calibrated point; additional calibration above 30 dB would be needed for a definitive 8×8 decision. The platform runs in floating-point MATLAB with no real-time implementation.

14. Conclusion and Future Work

Section 14 concludes from the stored files. The practical delayed-EESM controller maintained the 0.10 BLER target across the sweep on both arrays: it reached the target at 12 dB and held measured BLER at or below it at every subsequent point, with the worst compliant value at the crossing itself, 0.075 on the 8×8 array and 0.034 on the 16×16 array, and compliant-region averages of 0.025 and 0.014. With mapping-matched curves the mean and EESM controllers are now a controlled comparison, both compliant after their crossings, with the mean controller reaching a slightly higher terminal goodput by climbing to a higher average CQI. Against the Project 1 threshold adaptation the receiver-driven controller gained 9.9 percent goodput at 30 dB on the 8×8 array and 30.0 percent on the 16×16 array while replacing near-total target violation with unbroken compliance, and through the middle of both sweeps it delivered useful goodput where the threshold rule delivered almost none. The one-slot CQI delay cost 333 bits per slot at 30 dB against the same-slot bound on the 8×8 array and 1.25 percent on the 16×16 array, and the practical controller finished within 1.0 percent of the pilot-time-CSI reference at 8×8 and within 1.4 percent at 16×16. The outer loop was stable and convergent on both arrays, settled a positive offset of about 0.8 and 1.2 dB, and held the running BLER just below the 0.10 target rather than far under it, which on the 16×16 array produced the highest terminal goodput of any method, 125860 bits per slot, above the pilot-time-CSI reference because that reference bounds only the reporting loss. The two arrays differed as the calibration predicted: both complied from 12 dB onward, the 16×16 array selected slightly higher entries through the transition band and gained the MCS 4 entry that the 8×8 calibration did not admit, and its terminal goodput of 106457 bits

per slot exceeded the 8×8 figure of 52341 by a factor of 2.03, matching the doubling of the spatial streams.

Future work follows the limitations: LDPC coding with standardized segmentation, HARQ with soft combining, standardized CQI and MCS tables, rank adaptation and per-layer MCS, multi-user scheduling on the shared calibrated curves, fixed-point C implementation of the controller, and a real-time demonstration of the adaptation loop.

15. References

- [1] A. J. Goldsmith and S.-G. Chua, "Variable-rate variable-power MQAM for fading channels," *IEEE Transactions on Communications*, vol. 45, no. 10, pp. 1218–1230, 1997.
- [2] S. T. Chung and A. J. Goldsmith, "Degrees of freedom in adaptive modulation: a unified view," *IEEE Transactions on Communications*, vol. 49, no. 9, pp. 1561–1571, 2001.
- [3] E. Dahlman, S. Parkvall, and J. Sköld, *5G NR: The Next Generation Wireless Access Technology*, 2nd ed. Academic Press, 2020.
- [4] A. Ghosh, J. Zhang, J. G. Andrews, and R. Muhamed, *Fundamentals of LTE*. Prentice Hall, 2010.
- [5] Ericsson, "System-level evaluation of OFDM — further considerations," 3GPP TSG-RAN WG1 R1-031303, 2003.
- [6] K. Brueninghaus, D. Astély, T. Sälzer, S. Visuri, A. Alexiou, S. Karger, and G.-A. Seraji, "Link performance models for system level simulations of broadband radio access systems," in *Proc. IEEE PIMRC*, 2005, pp. 2306–2311.
- [7] J. C. Ikuno, M. Wrulich, and M. Rupp, "System level simulation of LTE networks," in *Proc. IEEE VTC-Spring*, 2010.
- [8] A. Sampath, P. S. Kumar, and J. M. Holtzman, "On setting reverse link target SIR in a CDMA system," in *Proc. IEEE VTC*, 1997, pp. 929–933.
- [9] D. L. Goeckel, "Adaptive coding for time-varying channels using outdated fading estimates," *IEEE Transactions on Communications*, vol. 47, no. 6, pp. 844–855, 1999.
- [10] S. Catreux, V. Erceg, D. Gesbert, and R. W. Heath, "Adaptive modulation and MIMO coding for broadband wireless data networks," *IEEE Communications Magazine*, vol. 40, no. 6, pp. 108–115, 2002.
- [11] M. M. Rahman, "Design, verification, and comparative performance evaluation of a scalable 8×8 and 16×16 MIMO-OFDM PHY simulator with threshold-based adaptive modulation and fixed-rate channel coding," technical report, 2026.
- [12] S. Lin and D. J. Costello, *Error Control Coding*, 2nd ed. Prentice Hall, 2004.
- [13] J. Hagenauer, "Rate-compatible punctured convolutional codes (RCPC codes) and their applications," *IEEE Transactions on Communications*, vol. 36, no. 4, pp. 389–400, 1988.
- [14] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.